

WHO PAYS WHEN? THE INTERGENERATIONAL INCIDENCE OF THE GREEN TRANSITION

BARBARA ANNICCHIARICO JEAN-GUILLAUME SAHUC GAUTHIER VERMANDEL

ABSTRACT. Who pays for the green transition depends on when we look. We quantify how carbon pricing redistributes welfare across generations using an overlapping-generations model of the euro area. Carbon pricing generates a *redistribution force* through two opposing channels: a labor-income channel, driven by wage compression, and a financial-wealth channel, driven by asset revaluation and higher real returns. As life-cycle exposure and the relative strength of these channels evolve, so does the identity of winners and losers. Initially, working-age households lose up to 5 percent of permanent consumption, while older households gain similarly; this ranking later reverses as climate benefits materialize. Following generations over their lifetimes reveals a *demographic footprint*, with the 2020 cohort bearing the largest lifetime loss, at 5.5 percent. Finally, carbon-revenue recycling generates a *distributional third dividend*: aggregate welfare gains can be shared across age groups while reducing consumption inequality, although some generations face deeper lifetime losses.

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Keywords: Carbon pricing, intergenerational redistribution, green transition, overlapping generations, revenue recycling.

B. Annicchiarico: Roma Tre University, Via Ostiense, 163–00154 Roma, Italy (e-mail: barbara.annicchiarico@uniroma3.it); J.-G. Sahuc: Banque de France, 31 rue Croix des Petits Champs, 75049 Paris, France (e-mail: jean-guillaume.sahuc@banque-france.fr); G. Vermandel: Ecole Polytechnique, CMAP UMR CNRS 7641, Route de Saclay, Palaiseau and University Paris-Dauphine, LEDA UMR CNRS 8007, Place du Maréchal de Lattre de Tassigny, 75016 Paris, France (e-mail: gauthier@vermandel.fr). We thank Marcin Bielecki, Rebecca Clipal, Janusz Jablonowski, Marcin Kolasa, and participants to several conferences for their comments. The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Banque de France or the Eurosystem. Gauthier Vermandel thanks the Stress-test chair for financial support. Declarations of interest: none.

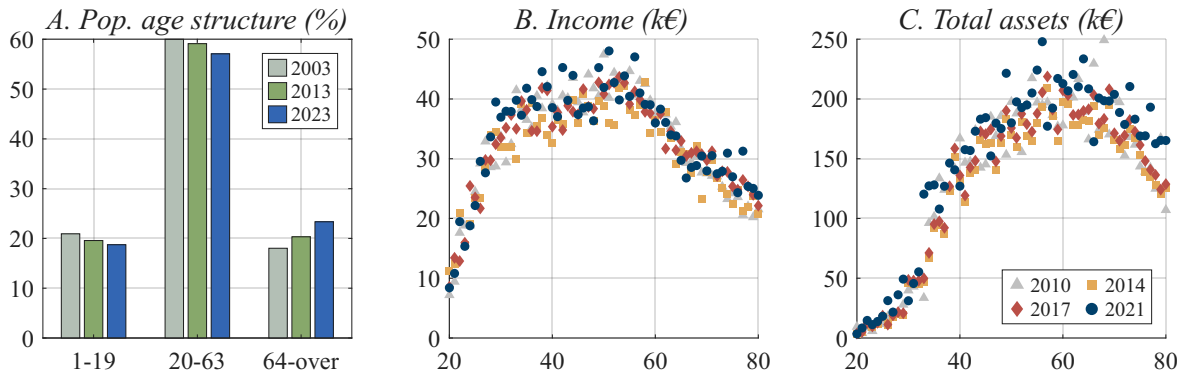
1. INTRODUCTION

Who pays for the green transition depends on when we ask. Carbon pricing unfolds over decades, long enough for today’s workers to become tomorrow’s retirees. Two clocks therefore run simultaneously: households age and shift from labor earnings toward accumulated wealth, while the economic effects of climate policy evolve from the costs of mitigation toward the benefits of lower climate damages. Their interaction implies that the same generation can lose in one phase of the transition and gain in another. Yet the intergenerational incidence of climate policy is typically assessed either at a point in time or by comparing lifetime welfare across generations. The first provides a snapshot of incidence; the second summarizes its lifetime consequences. Neither reveals the path connecting the two: how winners and losers change over time and how successive exposures accumulate across the life cycle. This distinction matters for policy design: compensating those who lose today may fail to compensate the generations that bear the largest costs over the transition.

We bring these perspectives together by defining the *redistribution force* of climate policy as the joint action of two opposing channels. By raising production costs during the mitigation phase, carbon pricing compresses wages and disproportionately affects households dependent on labor income – the labor-income channel. At the same time, weaker aggregate demand dampens inflation, raising the real value of nominal assets and real financial returns – the financial-wealth channel. Our contribution is to characterize how this redistribution force evolves as households age, mapping successive policy exposures into cumulative welfare consequences across cohorts. The euro area is particularly well suited to this analysis, given its rapid demographic change and pronounced life-cycle heterogeneity in household balance sheets. Figure 1 documents these features: the population is aging rapidly, while income composition and asset holdings vary sharply over the life cycle. Meeting the Paris Agreement, meanwhile, requires a sustained adjustment in carbon prices and abatement extending over several decades. The economic transition and the household life cycle therefore unfold over comparable horizons.

To quantify the *redistribution force*, we build an overlapping-generations macro-climate model of the euro area that combines the life-cycle heterogeneity needed to measure incidence with the long-run economic and climate dynamics of the transition. On the household side, Household Finance and Consumption Survey data discipline age profiles of labor income, asset holdings, portfolio composition, and consumption responses, which determine households’ exposure to the labor-income and financial-wealth channels. On the transition

FIGURE 1. Population structure and household balance sheets in the euro area



Note: This figure reports the population age structure by major age groups as a share of total population (Panel A), median before-tax income (Panel B), and total assets (Panel C) by single year of age for households aged 20 to 80. Data are drawn from the United Nations World Population Prospects (United Nations, 2024) and four waves of the Household Finance and Consumption Survey (HFCN, 2023).

side, a DICE-type climate block and long-run trends in population, productivity, carbon intensity, and abatement efficiency allow mitigation costs and climate damages to evolve over a decades-long horizon. Solving the model nonlinearly along the full transition path allows us to measure intergenerational incidence in two complementary ways: cross-sectionally, by identifying who gains and loses at each date, and longitudinally, by following birth cohorts and cumulating welfare effects over their remaining lifetimes.

Three findings emerge. First, the *redistribution force* generates a pronounced but shifting pattern of gains and losses across age groups. During the initial phase of the transition, the labor-income channel dominates: working-age households incur welfare losses of about 5 percent of permanent consumption, while retirees gain by a similar magnitude due to the financial-wealth channel. These differences initially widen before progressively narrowing and eventually reversing as wage compression recedes, financial gains dissipate, and climate benefits materialize. Who bears the transition therefore changes over time. Second, following birth cohorts over their lifetimes reveals a persistent *demographic footprint* that successive cross-sections conceal. Cohorts retiring near the beginning of the transition have already accumulated wealth and largely avoid subsequent labor-income losses. Cohorts whose working lives overlap with the costly phase instead experience wage compression during their main earning years, which reduces consumption and saving, and leaves them with lower accumulated wealth later in life. These effects persist beyond direct labor-income exposure and generate substantial differences in lifetime welfare across generations. In our baseline calibration, the generation born in 2020 bears the largest lifetime welfare loss, equivalent to about 5.5 percent of permanent consumption. Third, carbon-revenue recycling can extend

the conventional double dividend to a *distributional third dividend*. Labor-income tax cuts direct more of the welfare gains toward workers, whereas financial-income tax cuts provide greater compensation to retirees and stimulate investment. Combining the two instruments, we identify a policy range in which aggregate welfare gains are shared across working-age and retired households while consumption inequality is reduced. In our baseline calibration, this requires allocating 40 to 68 percent of carbon revenues to financial-income tax relief, with the remainder devoted to labor-income tax cuts. Yet this third dividend does not eliminate the *demographic footprint*: some generations whose working lives overlap most strongly with the costly phase of the transition remain worse off over their lifetimes.

This paper contributes to two strands of literature on the distributional effects of climate policy. First, we contribute to the literature on the intergenerational incidence of climate policy in overlapping-generations economies (Karp and Rezai, 2014; Kotlikoff et al., 2021, 2024; Fried et al., 2024; Kuhn and Schlattmann, 2024; Rezai and van der Ploeg, 2026). This literature has established that the costs and benefits of climate policy are unevenly distributed across generations, reflecting differences in remaining lifetimes, exposure to climate damages, and the timing of mitigation costs.¹ Recent work has enriched this analysis by incorporating richer household heterogeneity and transition dynamics. Our contribution is to characterize intergenerational incidence along the full transition path, jointly accounting for calendar time and life-cycle time. We discipline the life-cycle heterogeneity underlying this incidence with euro-area micro data, using detailed age profiles of labor income, asset holdings, and survival. This combination allows us to quantify how the identity of winners and losers changes along the transition and to uncover a persistent demographic footprint on lifetime welfare that successive cross-sectional measures of incidence can miss.

Second, our paper contributes to the growing literature studying climate policy in heterogeneous agent economies. This literature has shown that the incidence of carbon pricing depends critically on heterogeneity in income, wealth, consumption patterns, financial constraints, and exposure to climate damages (Benmir and Roman, 2022; Boehla and Budianto, 2024; Ascari et al., 2026; Belfiori et al., 2026; Douenne et al., 2026; Labrousse and Perdereau, 2026). These models uncover substantial differences in welfare effects across households and highlight the roles of labor-income exposure, asset positions, and fiscal recycling in shaping

¹The intergenerational implications of environmental taxation have been studied since the double dividend debate. Building on Baumol and Oates (1988), overlapping-generations frameworks (John and Pecchenino, 1994; Bovenberg and Heijdra, 1998) established that environmental taxation is generally not welfare neutral across cohorts.

distributional outcomes. Our paper adds a distinct dimension of heterogeneity: age. Existing heterogeneous-agent models characterize redistribution across households with different economic states, but do not distinguish households by their position in the life cycle. We instead study redistribution across generations that coexist at different ages and therefore differ systematically in their dependence on labor income and accumulated financial wealth. As households age, the relative importance of these two sources of income changes, altering their exposure to the distributional forces generated by carbon pricing. Introducing age therefore allows us to move from heterogeneity across households to intergenerational heterogeneity and to characterize how climate policy redistributes welfare across generations as they move through the transition.

The remainder of the paper is organized as follows. Section 2 presents the model, and Section 3 describes its calibration to euro-area data. Section 4 characterizes the aggregate effects of the green transition and the redistribution force across age groups. Section 5 follows generations through the transition and documents its demographic footprint. Section 6 examines how carbon-revenue recycling reshapes intergenerational redistribution and inequality. Section 7 concludes.

2. A MACRO-CLIMATE MODEL WITH OVERLAPPING GENERATIONS

The economy is described by a macro-climate model with overlapping generations of finitely lived households. Households enter the model when they start working at a real-world age of 20 (index $j = 1$), live up to age 99 ($J = 80$) and survive each period (a year) with an age-specific probability $1 - \omega_j$ (Bielecki et al., 2022, 2023). They work until the retirement age of 64 ($JR = 45$). As a result, the economy is populated by 80 overlapping cohorts of varying size in each period. Households maximize intertemporal utility by choosing consumption, labor supply, and financial asset holdings. On the production side, perfectly competitive final-good firms combine differentiated intermediate goods produced by monopolistically competitive firms that face Rotemberg price adjustment costs. Investment funds intermediate household savings into portfolios of physical capital and bonds and rent capital services to intermediate-good producers. Production generates CO_2 emissions, whose accumulated stock affects productivity and therefore alters production possibilities (Golosov et al., 2014). Firms do not internalize the environmental damages associated with their emissions, which motivates the implementation of a carbon tax by the government. Firms can partially reduce emissions through costly abatement activities (Heutel, 2012; Annicchiarico and Di Dio, 2015; Jondeau et al., 2026). The economy is further characterized by monetary and fiscal authorities

conducting stabilization and redistribution policies. Finally, the model incorporates four exogenous long-run trends (population dynamics, carbon intensity, abatement efficiency, and technological progress) to capture the structural evolution of the economy and emissions during the transition. Accounting for these structural trends is important in macro-climate models because they shape long-run emissions trajectories, climate outcomes, and therefore equilibrium expectations during the transition.

2.1. Household sector.

2.1.1. *Households' preferences.* Let $N_{j,t}$ denote the mass of households of the generation j alive at time t . The representative household of the generation j derives utility from consumption, $c_{j,t}$, and disutility from labor $h_{j,t}$. The expected lifetime utility function is of the form:

$$\mathcal{U}_{j,t} = \mathbb{E}_t \left\{ \sum_{s=0}^{J-j} \beta^s \frac{N_{j+s,t+s}}{N_{j,t}} \left(\log c_{j+s,t+s} - \varphi_{j+s} \frac{h_{j+s,t+s}^{1+\nu}}{1+\nu} \right) \right\}, \quad (1)$$

where $\mathbb{E}_t$ denotes the mathematical expectation operator conditional upon information available at t , $\beta \in (0, 1)$ is the constant discount factor, the ratio $N_{j+s,t+s}/N_{j,t}$ measures the probability of staying alive in other s years, φ_{j+s} is an age-dependent scale coefficient that measures the disutility of labor and $\nu > 0$ is the inverse of the Frisch labor supply elasticity. Households work only until retirement age JR : for $j \geq JR$ we set $h_{j,t} = 0$, so retirees derive utility from consumption alone.

The period-by-period budget constraint of the typical household j reads as

$$\begin{aligned} P_t(c_{j,t} + a_{j+1,t+1} + \mathcal{C}_{j,t}^x) &= (1 - \tau_h) \mathcal{I}_{j < JR} W_t z_j h_{j,t} \\ &+ (1 - \tau_a)(R_t^a - 1)P_{t-1}a_{j,t} + P_{t-1}a_{j,t} + P_t beq_t + P_t \tau_{j,t}, \end{aligned} \quad (2)$$

where P_t is the price index and $a_{j,t}$ denotes the real stock of financial wealth carried over from the previous period.² Financial wealth yields a nominal gross return R_t^a (where τ_a is the tax rate on financial income), τ_h is the tax rate on labor income, $\mathcal{I}_{j < JR}$ is an indicator variable that equals 1 if the household is of working age, W_t is the nominal wage and z_j measures an age-dependent labor productivity level. In addition, two boundary conditions close the household's problem. First, members of the youngest generation enter adulthood with no financial assets, that is $a_{1,t} = 0$. Second, as agents do not survive beyond age J , they consume

²The notation $a_{j+1,t+1}$ reflects the timing convention of the model: it denotes the financial wealth accumulated by a generation- j household during period t , which becomes available in period $t + 1$, when that household transitions to age $j + 1$. Thus, assets carried into period $t + 1$ are indexed by the next-period age and the next-period date.

all remaining wealth in their final period of life, implying $a_{j+1,t+1} = 0$. However, since most agents pass away before reaching their maximum life expectancy, they leave unintentional bequests of financial wealth that are confiscated by the fiscal authority and then redistributed equitably among all surviving agents through lump-sum transfers, denoted beq_t . The term $\tau_{j,t}$ is a catch-all variable that includes all net transfers made by the fiscal authority. Together, these elements determine the household's net cash flow,

$$x_{j,t} = (1 - \tau_h) \mathcal{I}_{j < JR} \frac{W_t}{P_t} z_j h_{j,t} + beq_t + \tau_{j,t} - c_{j,t}, \quad (3)$$

which is the residual that accumulates into financial wealth. Rebalancing wealth is not frictionless: the household pays a quadratic cost on the size of this flow,

$$c_{j,t}^x = \frac{\kappa_{x,t}}{2} x_{j,t}^2, \quad \kappa_{x,t} > 0. \quad (4)$$

The friction makes any deviation from a hand-to-mouth allocation costly, and is meant to capture the degree of illiquidity of financial wealth.³ The friction binds in steady state as well as along transitions: working-age cohorts pay a non-zero cost on the saving needed to fund retirement, and retirees pay a cost on dissaving.

Finally, as in [Huggett \(1996\)](#), households face a no-borrowing constraint:

$$a_{j,t} \geq 0 \quad \text{for all } j \text{ and } t. \quad (5)$$

This constraint is the key ingredient that generates a realistic life-cycle profile of marginal propensities to consume, in line with the empirical evidence in [Christelis et al. \(2019\)](#). Without it, young households would smooth transitory income shocks by borrowing against future earnings, producing a counterfactually low marginal propensity to consume at the start of the life cycle.⁴

2.1.2. Demographics and aggregation. The demographic structure is determined by the growth rate of the youngest generation, $n = N_{1,t+1}/N_{1,t} - 1$, assumed exogenous and constant, and by the age-dependent mortality risk ω_j , assumed time-invariant. The total number of agents

³In the spirit of [Kaplan and Violante \(2014\)](#), the adjustment cost limits households' ability to re-balance their asset positions and hence to smooth consumption, raising the marginal propensity to consume out of transitory income fluctuations. To preserve stationarity when the model is expressed in efficiency units, the parameter governing the size of this cost scales with trend productivity, Z_t : $\kappa_{x,t} = \kappa_x / Z_t$, where $\kappa_x > 0$.

⁴This is the case, for instance, in [Bielecki et al. \(2022\)](#), who abstract from a borrowing limit, which tends to make the youngest cohorts exhibit very low (close to zero) marginal propensities to consume, at odds with the data. In our context, this friction is crucial: young households are less able to smooth the (transition) costs induced by climate policies, so the green transition can generate stronger intergenerational asymmetries.

alive at time t is thus given by

$$N_t = \sum_{j=1}^J N_{j,t}, \quad (6)$$

where the mass of agents in each cohort $j > 1$ evolves over time as follows

$$N_{j+1,t+1} = (1 - \omega_j) N_{j,t}. \quad (7)$$

We assume the economy starts in the demographic steady state, so that total population grows at the constant rate n while the absolute size of each cohort grows over time. We therefore work with relative cohort sizes,

$$N_j^{rel} = \frac{N_{j,t}}{N_t}, \quad (8)$$

which are time-invariant given our assumptions on n and ω_j . Equation (7) can then be rewritten in stationary form as

$$N_{j+1}^{rel} = \frac{1 - \omega_j}{1 + n} N_j^{rel}. \quad (9)$$

Asset markets clear at the end of period t , before period- $(t + 1)$ mortality is realized. Aggregate wealth carried into $t + 1$ therefore includes the assets of cohorts that will not survive; their balances are redistributed in the following period through the bequest term beq_{t+1} . The aggregate allocations over all existing households can then be represented in per capita terms as follows:

$$c_t = \sum_{j=1}^J N_j^{rel} c_{j,t}, \quad (10)$$

$$h_t = \sum_{j=1}^{JR-1} N_j^{rel} z_j h_{j,t}, \quad (11)$$

$$a_{t+1} = \sum_{j=1}^{J-1} \frac{N_j^{rel}}{1 + n} a_{j+1,t+1}, \quad (12)$$

$$C_t^x = \frac{\kappa_{x,t}}{2} \sum_{j=1}^J N_j^{rel} x_{j,t}^2, \quad (13)$$

$$\tau_t = \sum_{j=1}^J N_j^{rel} \tau_{j,t}. \quad (14)$$

Finally, the unintentional bequests left by deceased agents rebated to surviving households are given by

$$beq_t = \sum_{j=2}^J \frac{\omega_{j-1} N_{j-1}^{rel}}{1 + n} \frac{[1 + (1 - \tau_a)(R_t^a - 1)] a_{j,t}}{\pi_t}, \quad (15)$$

where $\pi_t = P_t/P_{t-1}$ is gross inflation, while the term $\omega_{j-1}N_{j-1}^{rel}$ is the relative mass of cohort $j - 1$ that died at the end of period $t - 1$.

2.2. Business sector. The business sector consists of three types of agents: final-good producers, intermediate-goods producers, and investment funds.

2.2.1. Final good producers. At every point in time t , a perfectly competitive sector produces a final good y_t by combining a continuum of intermediate goods $y_{i,t}$, $i \in [0, N_t]$, according to the technology $y_t = \left(\frac{1}{N_t} \int_0^{N_t} y_{i,t}^{1/\varepsilon_p} di \right)^{\varepsilon_p}$, where $\varepsilon_p > 1$ is the (gross) markup parameter. Equivalently, the elasticity of substitution across varieties is $\varepsilon_p/(\varepsilon_p - 1)$, so a higher ε_p corresponds to less substitutable varieties and a larger desired markup. As the population grows, the measure of available intermediate goods expands accordingly, reflecting an increasing mass of variety in production.⁵ Final good producing firms take their output price, P_t , and their input prices, $P_{i,t}$, as given and beyond their control. Profit maximization implies $y_{i,t} = (P_{i,t}/P_t)^{-\frac{\varepsilon_p}{\varepsilon_p-1}} y_t$ from which we obtain the relationship between the price of the final good and the prices of intermediate goods $P_t \equiv \left(\frac{1}{N_t} \int_0^{N_t} P_{i,t}^{\frac{1}{1-\varepsilon_p}} di \right)^{1-\varepsilon_p}$.

2.2.2. Intermediate-goods firms. Intermediate good i is produced by a monopolistically competitive firm with the following technology:

$$y_{i,t} = (1 - \mathcal{D}(T_t)) (k_{i,t}^s)^\alpha (Z_t h_{i,t})^{1-\alpha}, \quad (16)$$

where $\mathcal{D}(T_t)$ is a damage function capturing the impact of climate change on production (specified below), $k_{i,t}^s$ denotes the flow of capital services and $h_{i,t}$ the labor input. The parameter $\alpha \in (0, 1)$ measures the output elasticity with respect to capital and Z_t denotes labor productivity. It follows the process $Z_t = (1 + g_{z,t})Z_{t-1}$, where $g_{z,t} = g_{z,t-1}(1 - \delta_z)$ is the productivity growth rate and $\delta_z \geq 0$ is the rate of decline in the productivity growth rate.

During production, firm i generates CO₂ emissions $e_{i,t}$ according to:

$$e_{i,t} = \sigma_t (1 - \mu_{i,t}) y_{i,t}, \quad (17)$$

where $\mu_{i,t} \in [0, 1]$ is the abatement effort (i.e., the fraction of emissions abated), $\sigma_t > 0$ represents the emissions per unit of output in absence of any reduction effort and is assumed to decay over time at rate $\delta_\sigma \geq 0$, thus $\sigma_t = \sigma_{t-1}(1 - \delta_\sigma)$. Firms may substitute carbon-intensive

⁵The $1/N_t$ normalization inside the aggregator removes love-of-variety effects: a proportional expansion of varieties driven by population growth scales the economy up but does not by itself raise per-capita output. The mass of varieties available in production therefore grows in step with the population without generating endogenous productivity gains.

technologies with low-carbon technologies, but this change in the existing lines of production is costly. We assume that the cost of abatement technology (expressed in units of the final good) is given by:

$$C_{i,t}^{\mu} = \phi_{1,t} \mu_{i,t}^{\phi_2} y_{i,t}, \quad (18)$$

where $\phi_{1,t} > 0$, and $\phi_2 > 1$ are technological parameters that determine the cost of reducing emissions. In line with the polluter-pays principle, firms face an environmental policy requiring them to pay a carbon price on each unit of emissions, treated as exogenous. Abatement costs are assumed to decline over time at rate $\delta_{\phi} \geq 0$ due to exogenous technological progress, so that $\phi_{1,t} = \phi_{1,t-1}(1 - \delta_{\phi})$.

Intermediate firms set prices subject to Rotemberg adjustment costs:

$$C_{i,t}^p = \frac{\kappa_p}{2} \left(\frac{P_{i,t}}{P_{i,t-1}} - \pi \right)^2 y_t, \quad (19)$$

where $\kappa_p > 0$ is the price stickiness parameter and π is steady-state gross inflation. In period t , nominal profits of a representative firm i are given by

$$P_t d_{i,t} = P_{i,t} y_{i,t} - W_t h_{i,t} - R_t^k k_{i,t}^s - \tau_{e,t} e_{i,t} - P_t C_{i,t}^{\mu} - P_t C_{i,t}^p, \quad (20)$$

where $\tau_{e,t}$ denotes the exogenous carbon tax while R_t^k is the nominal rental rate of capital services. Thus, the problem faced by firms is that of choosing factor inputs, abatement effort, and their own price so as to maximize the present discounted value of current and expected profits

$$\mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \left(\prod_{\varepsilon=1}^s R_{t+\varepsilon}^{-1} \right) P_{t+s} d_{i,t+s} \right\}, \quad (21)$$

subject to demand from final firms and the production function. Intermediate goods firms are assumed to be risk neutral and thus discount future profits at the nominal risk-free rate R_t .

2.2.3. Investment funds. Perfectly competitive investment funds use households' savings to buy and manage a portfolio of assets, transferring every period the earned gross return back to households. The portfolio consists of physical capital k_t , bonds b_t (both expressed in per capita terms), and claims on intermediate-good producing firms' profits $d_t = (1/N_t) \int_0^{N_t} d_{i,t} di$. Investment funds rent capital services to intermediate-good producers. Capital services are defined as

$$k_t^s = u_t k_t, \quad (22)$$

where u_t denotes the capital utilization rate. Adjusting the capital utilization rate is costly and the cost is borne by investment funds. Following [Smets and Wouters \(2003, 2007\)](#), utilization entails the following convex cost (expressed in units of the final good):

$$C_t^u = \frac{1 - \psi_k}{\psi_k} r_k \left[\exp \left(\frac{\psi_k}{1 - \psi_k} (u_t - 1) \right) - 1 \right] k_t, \quad (23)$$

where $\psi_k \in (0, 1)$ governs the curvature of the cost around the steady state and r_k is the steady-state real rental rate of capital.

A representative investment fund operates to maximize the anticipated present value of future gross returns:

$$\mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \left(\prod_{\varepsilon=1}^s R_{t+\varepsilon}^{-1} \right) \left[\left(R_{t+s}^k u_{t+s} + (1 - \delta) Q_{t+s} \right) k_{t+s} - P_{t+s} C_{t+s}^u \right. \right. \\ \left. \left. + R_{t-1+s} P_{t-1+s} b_{t+s} + P_{t+s} d_{t+s} \right] \right\} \quad (24)$$

where δ is the capital depreciation rate, and Q_t is the nominal price of a unit of capital. Risk neutrality justifies discounting at the risk-free rate R_t . As in [Bielecki et al. \(2023\)](#), investment outlays do not appear in the objective: they are financed out of new household savings through the balance-sheet identity below, and only the flows rebated to households enter the maximization.

The nominal balance sheet of investment funds at the end of period t is

$$(1 + n) P_t a_{t+1} = Q_t (1 - \delta) k_t + P_t i_t + (1 + n) P_t b_{t+1}, \quad (25)$$

where i_t denotes investment in physical capital. Physical capital in per capita terms is assumed to accumulate as follows:

$$(1 + n) k_{t+1} = (1 - \delta) k_t + \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t}{(1 + g_{z,t}) i_{t-1}} - 1 \right)^2 \right] i_t, \quad (26)$$

where $\kappa_i > 0$.

Since investment funds are risk-neutral, their portfolio choices imply equalization of ex-ante rates of return on capital and bonds:

$$\mathbb{E}_t \left\{ \frac{u_{t+1} R_{t+1}^k - P_{t+1} C_{t+1}^u / k_{t+1} + (1 - \delta) Q_{t+1}}{Q_t} \right\} = R_t. \quad (27)$$

Because all revenues are passed through to households, the ex-post nominal gross return on financial wealth R_t^a is implicitly defined by

$$R_t^a P_{t-1} a_t = \left[u_t R_t^k + (1 - \delta) Q_t \right] k_t - P_t C_t^u + R_{t-1} P_{t-1} b_t + P_t d_t. \quad (28)$$

Equation (28) closes the household-fund block: it pins down R_t^a as the residual gross return that exhausts the investment fund's revenues, and is the rate at which households' financial wealth a_t accumulates in the budget constraint (2).

2.3. Public sector. The government collects tax revenue from labor income, financial income, and emissions; spends on public consumption; pays transfers to households; and finances any residual through debt. Its flow budget constraint is

$$(1 + n) P_t b_{t+1}^g = R_{t-1} P_{t-1} b_t^g + P_t g_t + P_t \tau_t - \tau_{h,t} W_t h_t - \tau_{a,t} (R_t^a - 1) P_{t-1} a_t - \tau_{e,t} e_t, \quad (29)$$

where b_t^g is the stock of public debt in per capita terms carried from the previous period, g_t denotes public consumption and is assumed to evolve linearly with aggregate output. The term τ_t is the aggregate per-capita transfer defined in equation (14), and is set period by period to close the government budget constraint: transfers adjust residually so that the government runs a zero deficit at all times. The term e_t is the aggregate emissions per capita:

$$e_t = \frac{1}{N_t} \int_0^{N_t} e_{i,t} di. \quad (30)$$

In what follows, carbon tax revenues, $\tau_{e,t} e_t$, are either rebated to households as lump-sum transfers or used to reduce labor and/or financial income taxes.

The monetary authority follows a Taylor-type rule by gradually adjusting the nominal interest rate in response to inflation and output growth:

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R} \right)^{\varphi_r} \left[\left(\frac{\pi_t}{\pi} \right)^{\varphi_\pi} \left(\frac{y_t}{(1 + g_{z,t}) y_{t-1}} \right)^{\varphi_y} \right]^{1 - \varphi_r}, \quad (31)$$

where $\varphi_r \in [0, 1)$ governs interest-rate smoothing and $\varphi_\pi, \varphi_y > 0$ govern the responses to inflation and output growth.

2.4. Market clearing conditions, emissions and climate block. The equilibrium in the final goods market is given by

$$y_t = c_t + i_t + g_t + C_t^p + C_t^h + C_t^x + C_t^u, \quad (32)$$

where all variables are expressed in per capita terms.

The market-clearing conditions for capital and labor are

$$k_t = \frac{1}{N_t} \int_0^{N_t} k_{i,t} di \quad \text{and} \quad h_t = \frac{1}{N_t} \int_0^{N_t} h_{i,t} di, \quad (33)$$

and the aggregate net supply of bonds is equal to the level of government debt:

$$b_t = b_t^s, \quad (34)$$

where lowercase letters refer to per capita values.

Since at the optimum each firm undertakes the same abatement effort, domestic per-capita emissions are determined endogenously as

$$e_t = \sigma_t (1 - \mu_t) y_t, \quad (35)$$

where $E_t = N_t e_t$ denotes aggregate domestic emissions and μ_t is itself determined endogenously by firms optimizing against the carbon price.

We now turn to the climate block. Following [Barrage and Nordhaus \(2024\)](#), we assume a quadratic damage function that depends on the (global) temperature anomaly:

$$\mathcal{D}(T_t) = \gamma_1 T_t + \gamma_2 T_t^2. \quad (36)$$

where γ_1 and γ_2 are the coefficients of the process.

To capture the transient climate response, we adopt a parsimonious linear mapping from global emissions to the temperature anomaly, consistent with the evidence reviewed by [Dietz et al. \(2021\)](#). Two variables are taken as exogenous from the DICE-2023 scenarios of [Barrage and Nordhaus \(2024\)](#): the path of total global emissions E_t^{tot} and the carbon tax $\tau_{e,t}$. Conditional on $\tau_{e,t}$, each firm optimally chooses its abatement effort μ_t , so that domestic emissions E_t are determined endogenously in general equilibrium. Rest-of-the-world emissions E_t^* are then simply deduced as the residual:

$$E_t^* = E_t^{tot} - E_t. \quad (37)$$

Temperature evolves according to

$$T_t = T_{t-1} + \zeta_T E_t^{tot}, \quad (38)$$

where $\zeta_T > 0$ governs the transient climate response. All the equilibrium conditions of the model are reported in [Appendix A](#).

3. CALIBRATION AND SOLUTION METHOD

We calibrate the model to the euro area at an annual frequency, taking 2023 as the reference initial date ($t_0 = 2023$).⁶ The calibration is disciplined to be consistent with both macroeconomic aggregates and household-level evidence. On the macro side, we target standard euro-area moments. On the micro side, we exploit age-specific moments from the HFCS to pin down life-cycle profiles of income and wealth, which are key to match cohort heterogeneity and age-dependent marginal propensities to consume.

3.1. Structural parameters. The structural parameters are calibrated using benchmark values from the literature and empirical estimates drawn from international databases. Table 1 summarizes the calibration. Panel A reports the climate block, largely based on [Barrage and Nordhaus \(2024\)](#), and includes parameters governing the temperature–emissions mapping, the damage function, abatement technology (level and curvature), and the trend in emissions intensity. The damage function $\mathcal{D}(T_t) = \gamma_1 T_t + \gamma_2 T_t^2$ follows the Nordhaus–Barrage specification: the linear coefficient is set to $\gamma_1 = 0$ and the quadratic coefficient to $\gamma_2 = 0.003467$, so that only the quadratic term is active at the calibrated values.

Panel B reports the private-sector parameters. The annual discount factor is set to $\beta = 0.99$, while the portfolio adjustment-cost curvature, $\kappa_x = 0.04$, is chosen to match marginal propensities to consume consistent with the evidence in [Christelis et al. \(2019\)](#). The inverse Frisch elasticity is $\nu = 2.21$, the capital share is $\alpha = 0.30$, and the annual depreciation rate is $\delta = 0.05$. The steady-state gross markup is $4/3$, and the investment adjustment-cost curvature is $\kappa_i = 5.28$. The Rotemberg price-adjustment parameter is set to $\kappa_p = 2.024$, obtained by applying the mapping in [Keen and Wang \(2007\)](#) to a quarterly price-adjustment frequency of 25 percent, as estimated by [Gautier et al. \(2024\)](#). Finally, steady-state gross inflation is set to $\pi = 1.02$.

Panel C reports the parameters governing fiscal policy and the monetary policy rule. Government purchases are set to 21 percent of output, based on data from the ECB Statistical Data Warehouse ([European Central Bank, 2024](#)) over the 1999–2023 period. The labor income tax rate, defined as income taxes as a share of gross wage earnings, is constructed as a GDP-weighted average across euro-area member states using [OECD \(2025\)](#). The financial income tax rate is similarly constructed as a weighted average across the same countries, using the effective tax rates on capital reported by [Bachas et al. \(2024\)](#). The monetary policy parameters are estimated from a log-linearized version of Equation (31) using euro-area data for

⁶This corresponds to the EA20 (20 member states after Croatia’s accession on 1 January 2023).

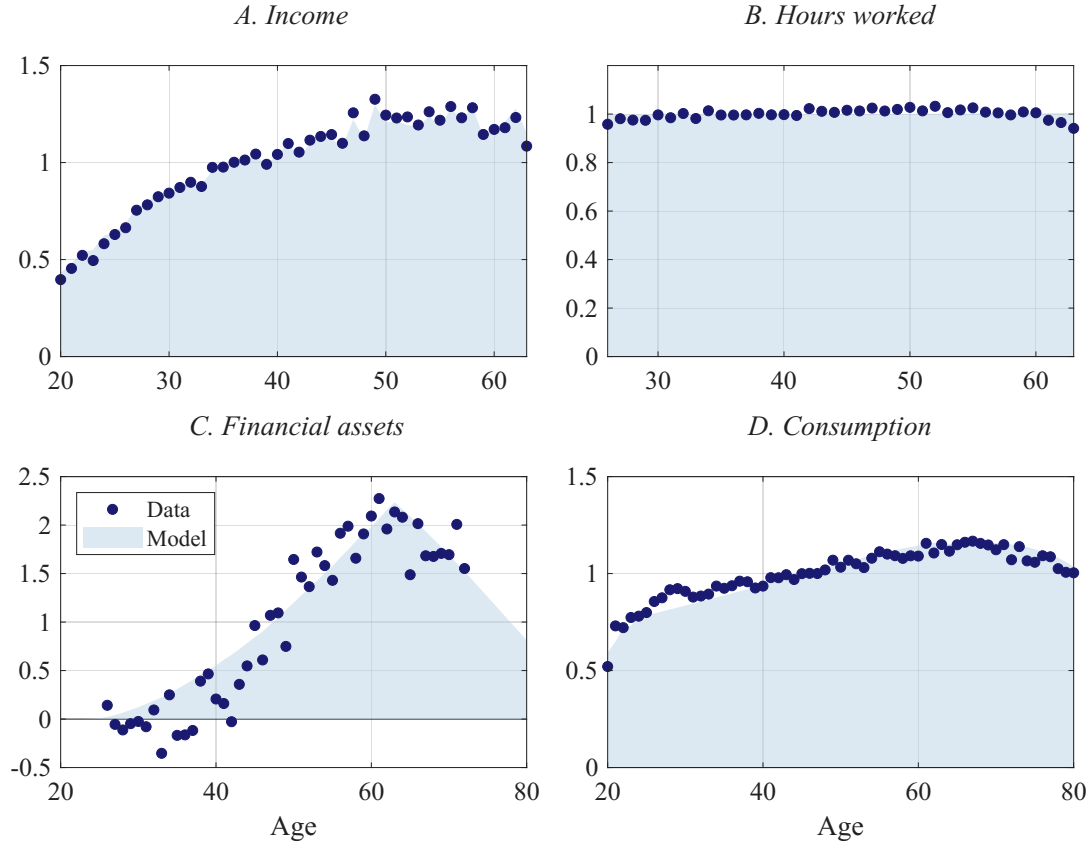
TABLE 1. Model parameters

Name	Symbol	Value	Sources
Panel A: Climate parameters			
Atmospheric conversion	ξ_T	5.4545×10^{-4}	Own calculations
Linear damage coefficient	γ_1	0	Barrage and Nordhaus (2024)
Quadratic damage coefficient	γ_2	0.003467	Barrage and Nordhaus (2024)
Abatement cost curvature	ϕ_2	2.600	Barrage and Nordhaus (2024)
Decay rate of abatement cost	δ_ϕ	0.017	Barrage and Nordhaus (2024)
Decay rate of emission intensity	δ_σ	0.000	Barrage and Nordhaus (2024)
Panel B: Structural parameters			
Population growth rate (%)	n	0.220	United Nations (2024)
Discount factor	β	0.990	Standard
Inverse of the Frisch elasticity of labor supply	ν	2.210	Mouabbi and Sahuc (2019)
Portfolio adjustment cost curvature	κ_x	0.040	Own calculation
Decay rate of productivity	δ_z	0.017	Barrage and Nordhaus (2024)
Capital depreciation rate	δ	0.050	European Central Bank (2006)
Capital share in output	α	0.300	McAdam and Willman (2013)
Investment adjustment cost curvature	κ_i	5.280	Mouabbi and Sahuc (2019)
Gross markup	ε_p	1.333	Christopoulou and Vermeulen (2012)
Price stickiness	κ_p	2.024	Gautier et al. (2024)
Panel C: Public sector parameters			
Share of government purchases in GDP	g/y	0.210	European Central Bank (2024)
Labor income tax	τ_h	0.178	OECD (2025)
Financial income tax	τ_a	0.337	Bachas et al. (2024)
Interest rate smoothing	φ_r	0.833	Own calculations
Reaction to inflation	φ_π	1.925	Own calculations
Reaction to GDP growth	φ_y	0.706	Own calculations
Gross inflation rate	π	1.020	European Central Bank's target

1990–2023 from an updated version of the Area Wide Model database (Fagan et al., 2005), aggregated to annual frequency.

3.2. Life-cycle profiles. We use individual data from the Household Finance and Consumption Survey hosted by the European Central Bank (https://www.ecb.europa.eu/stats/ecb_surveys/hfcs/html/index.en.html). The survey collects detailed information on household finances, including real assets and their financing, other liabilities and credit constraints, private businesses, financial assets, income, and consumption (see HFCN, 2023, Annicchiarico et al., 2025). The HFCS has been conducted in four waves (2010, 2014, 2017, and 2021) among households in all euro-area member countries. In the most recent wave, more than 83,000 households were interviewed. We average the results across the four waves to obtain estimates representative of roughly a decade. All monetary amounts are expressed in constant 2021 euros, deflated using the harmonised index of consumer prices (see Appendix B). We focus on three variables (HFCS code in parentheses): (i) labor income, defined as the sum of employee income (DI1100) and self-employment income (DI1200); (ii) hours

FIGURE 2. Life cycle profiles in the euro area



Notes: The panels show age profiles from the HFCS raw data (dots) and model-implied smoothed profiles (areas). All series are normalized by their average values over the working age (income and hours worked) or lifecycle (financial assets and consumption). Raw data were adjusted for household size and aggregated by cohort.

worked, defined as weekly hours worked in the main job (PE0600), set to zero for non-workers; and *iii*) financial assets, defined as the sum of deposits (DA2101), mutual funds (DA2102), bonds (DA2103), the value of non-self-employment private business (DA2104), publicly traded shares (DA2105), life insurance (DA2109), and business wealth (DA1200), minus the outstanding balances of mortgage (DL1100) and non-mortgage (DL1200) debt. These definitions are chosen to map the HFCS variables as closely as possible into the objects in the model. In particular, our measure of financial assets is a model-consistent net financial wealth concept and therefore differs from the broader measure of total assets reported in the introduction. While calibrating the model, we match the steady-state age profiles of labor income and hours worked exactly to their empirical counterparts by setting the age-specific productivity parameters z_j equal to the empirical ratio of labor income to hours worked.

Figure 2 presents the empirical and model-implied profiles. The income profile (Panel A) displays the canonical hump shape, peaking around ages 50-55 before declining toward

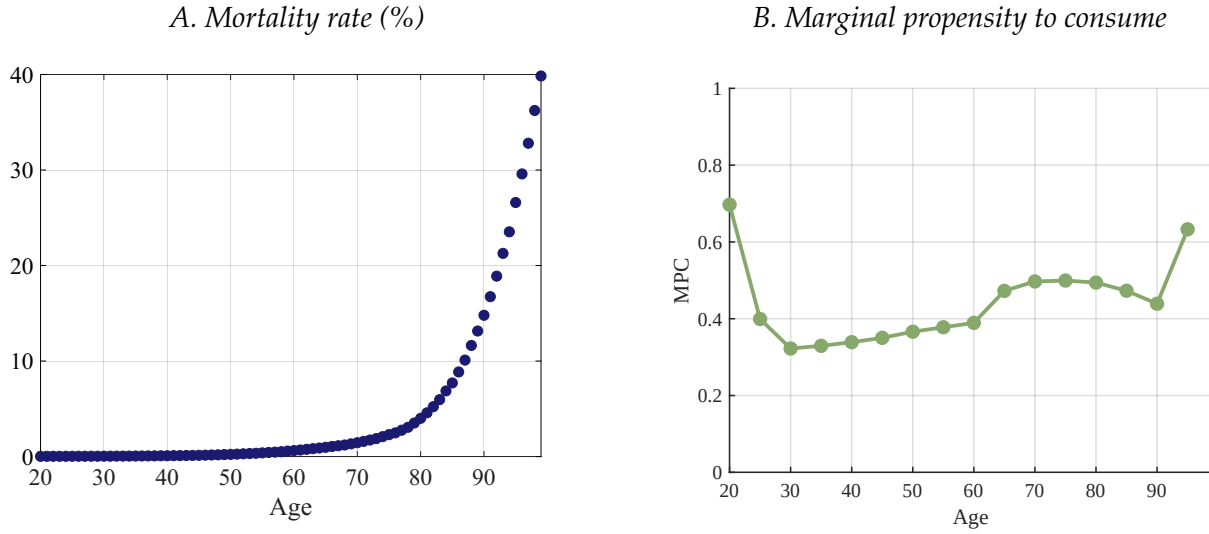
retirement, and the model tracks it closely. Hours worked (Panel B) remain relatively stable over most of the working life, so that the pronounced life-cycle variation in labor income primarily reflects the age profile of wages rather than changes in hours worked. Financial assets (Panel C) follow the textbook Modigliani pattern: negative net positions in the twenties reflecting borrowing, a steep accumulation phase through midlife, and a peak around age 60-65. The model reproduces the broad shape of the accumulation profile quite well despite abstracting from borrowing entirely, a restriction that mechanically rules out the negative asset positions observed among young households in the data yet leaves the overall fit largely intact. Consumption (Panel D) rises through most of the life cycle before declining slightly at older ages, reflecting life-cycle variation in household resources and consumption-smoothing behavior. Because the consumption profile is not directly targeted in the calibration, its close fit provides an additional validation of the model’s life-cycle mechanisms.

3.3. Demography. We use data from the 2024 revision of the United Nations’ World Population Prospects (<https://population.un.org/wpp/>) to calibrate population growth and age-dependent mortality risk. We assume a constant population structure, implying that the cohort aged 20 and the total population grow at the same rate. Age-specific mortality rates are constructed using the 2023 country-level Life Tables and corresponding cohort population data (Figure 3, Panel A).

3.4. Marginal propensity to consume over the life cycle. The model generates a pronounced life-cycle pattern of marginal propensities to consume (MPCs) that is central to the distributional effects of the transition. We simulate unexpected, transitory income shocks of +€1,000 and -€1,000 in 2024 and compute the corresponding impact MPCs for each age cohort, reporting their average in Figure 3, Panel B.⁷ MPCs are high at both ends of the life cycle for distinct reasons, implying imperfect aggregate insurance. Young households exhibit high MPCs because they hold little wealth and face a binding borrowing constraint. As a result, they behave hand-to-mouth: income shocks translate almost one-for-one into consumption. Negative shocks, such as wage compression from carbon pricing, are therefore expected to reduce consumption immediately and permanently. At older ages, MPCs rise due to end-of-life decumulation. As survival probabilities decline, the incentive to save weakens, and households consume a larger share of additional income. At intermediate ages, the model features

⁷For this experiment, the household budget constraint (2) is augmented with a cohort-specific transitory income shock $\epsilon_{j,t}$, so that the cash flow becomes $x_{j,t} = (1 - \tau_h)\mathcal{I}_{j < JR}(W_t/P_t)z_jh_{j,t} + beq_t + \tau_{j,t} + \epsilon_{j,t} - c_{j,t}$. The cohort-level MPC is the impact response of $c_{j,t}$ relative to the change in $\epsilon_{j,t}$. The shock is used only for this experiment and is set to zero elsewhere in the paper.

FIGURE 3. Mortality rate and marginal propensity to consume over the life cycle



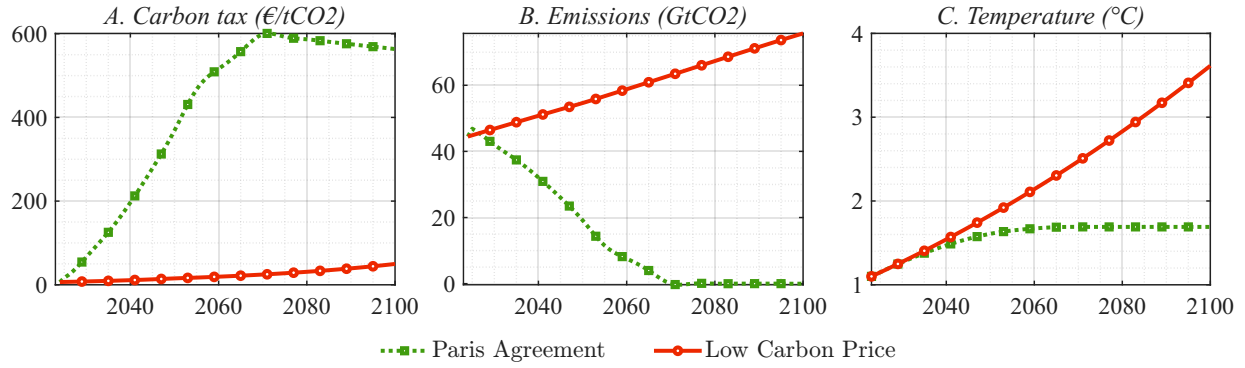
Notes: Panel A plots mortality rates in 2023 (United Nations, 2024). Panel B displays the marginal propensity to consume (MPC) by age, computed from the calibrated model following an unexpected transitory idiosyncratic income shock in 2024. The MPC is defined as the ratio of the contemporaneous consumption response to the income shock. For each age, the reported MPC is the average of the MPCs obtained from a +€1,000 and a -€1,000 income shock. The resulting age profile is smoothed over a five-year window.

wealthy hand-to-mouth households in the spirit of Kaplan and Violante (2014). These households hold illiquid assets but face short-run liquidity constraints due to portfolio adjustment costs, leading them to exhibit high MPCs despite positive wealth. This group is consistent with euro-area evidence from the HFCS and reinforces incomplete insurance, as even wealthy households respond strongly to income shocks. This MPC profile has two implications for the green-transition analysis that follows. First, aggregation is non-trivial: a representative-agent setup would average across cohorts and miss the heterogeneity. Second, any policy that redistributes income across age groups produces heterogeneous consumption responses, since cohorts with different MPCs absorb income changes to different extents.

3.5. Solution method and exogenous global climate paths. To solve the model, we rely on a perfect foresight solution method. The algorithm computes paths for endogenous variables consistent with the model's equations by solving the resulting simultaneous equation system via a Newton-type method, given a sequence of exogenous transition paths and the absence of aggregate uncertainty. This approach captures all nonlinearities of the model while remaining computationally tractable over the long horizons required by the transition analysis.

Our analysis relies on long-term projections generated by the model to explore alternative trajectories for the euro area up to 2100. Specifically, we assume that in 2023, agents receive

FIGURE 4. Exogenous global climate paths



Notes: The figure reports the exogenous carbon tax paths (in euros per ton of CO₂) and the resulting world emissions and temperature trajectories under the Paris Agreement ("R=1%") and Low Carbon Price ("Base") scenarios, taken from the DICE-2023 framework of [Barrage and Nordhaus \(2024\)](#).

information about a long-term carbon tax path. The Paris Agreement scenario implements a carbon tax consistent with limiting global warming to below 2°C above pre-industrial levels. The Low Carbon Price scenario extrapolates current policies as of 2023, with a global average carbon price of \$6/tCO₂ growing at 2.5% annually. These correspond to the "R=1%" and "Base" cases in [Barrage and Nordhaus \(2024\)](#), respectively.

We treat as exogenous both the carbon tax path (in euros per ton of CO₂) and the trajectory of global emissions, from which temperature is directly derived from Equation (38). Conditional on these inputs, euro-area abatement and domestic emissions are determined endogenously by firms optimizing against the carbon tax, while foreign emissions are obtained residually as the difference between global and euro-area emissions.

The model then jointly determines output, factor prices, inflation, and asset returns, which together shape the distributional incidence of the transition across cohorts. This strategy leaves the optimal determination of the carbon tax schedule and transition path outside the scope of the model, treating its trajectory as given by global climate commitments. Our focus is instead on tracing how a predetermined carbon tax path transmits through the euro-area economy and generates distinct welfare dynamics across cohorts. Figure 4 reports the exogenous carbon tax paths under the two scenarios, together with the corresponding global emissions and temperature trajectories.

4. THE DISTRIBUTIONAL INCIDENCE OF THE GREEN TRANSITION

The aggregate cost of the green transition masks a substantial redistribution across households of different ages. We characterize this redistribution by comparing the Paris-consistent

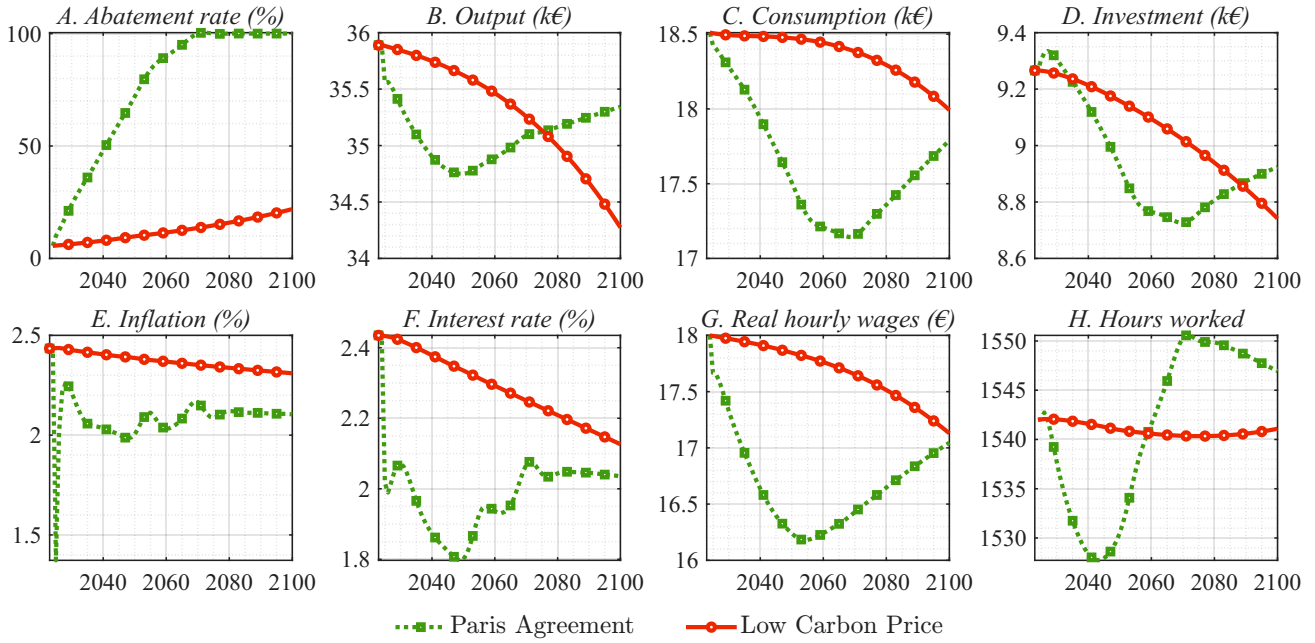
transition with the Low Carbon Price scenario. We first identify the aggregate forces generated by the transition and then map how their incidence across age groups evolves over time. The key question is whether the households that bear the transition's costs at its onset remain its losers as the transition unfolds.

4.1. Aggregate dynamics. The Paris-consistent transition requires a sustained increase in abatement effort over several decades. Firms progressively scale up abatement until emissions are nearly eliminated by the end of the century, whereas abatement remains negligible under low carbon pricing. Figure 5 reports the associated macroeconomic dynamics over 2023–2100.⁸ As we show below, the costs of this adjustment are concentrated in the early decades of the transition, while the benefits of lower climate damages materialize only gradually. Consistent with Sahuc et al. (2024) and Jondeau et al. (2026), the stronger carbon tax initially contracts economic activity, with output falling by about 3.5%. Higher production costs compress firm margins and reduce labor demand, lowering labor income and profits. The resulting decline in household disposable income depresses consumption, despite the fall in real interest rates, which in isolation would stimulate current spending. Consumption falls by about 7%, while investment declines by about 5% as firms respond to weaker demand and lower profitability. Although carbon pricing originates as a supply-side disturbance, its general-equilibrium effects resemble those of a negative demand shock: output, consumption, investment, and inflation all decline. The income-induced contraction in aggregate demand more than offsets the direct inflationary pressure from higher production costs. This demand amplification is central to the distributional effects of the transition, as lower labor income hurts workers while disinflation raises the real value of nominal wealth.

The disinflation is particularly pronounced at the onset of the transition, reflecting the forward-looking response to the announced carbon-price path. Under perfect foresight, households and firms immediately internalize future carbon-price increases, bringing forward the contraction in aggregate demand. Inflation therefore falls sharply on impact and remains below the Low Carbon Price path for several decades, as demand-side pressures dominate the direct inflationary effect of higher production costs. It gradually converges from around 2.4% in 2023 toward 2% by mid-century before partially recovering. Nominal interest rates decline in tandem, consistent with the monetary policy rule.

⁸In both scenarios, carbon tax revenues are returned equally to households via lump-sum transfers. All real variables are expressed in detrended per capita units, normalizing out population and productivity trend growth to isolate the purely transitional effects of climate policy.

FIGURE 5. Transition dynamics under alternative climate policy paths



Notes: The figure reports transition paths under the Paris Agreement ("R=1%", dashed green line) and Low Carbon Price ("Base", solid red line) scenarios, taken from the DICE-2023 framework of [Barrage and Nordhaus \(2024\)](#). All variables are expressed in detrended levels.

Over the longer run, the forces driving the transition reverse. As abatement costs recede and the capital stock adjusts, economic activity gradually recovers under the Paris Agreement. At the same time, lower emissions limit climate damages, whereas rising damages under the Low Carbon Price scenario increasingly weigh on productivity and output. The transition therefore involves a pronounced intertemporal trade-off: adjustment costs are borne early, while the economic benefits of mitigation materialize only gradually. This timing is central to the distributional analysis that follows, because different generations are exposed to these costs and benefits at different stages of their life cycle.

4.2. Who gains and who loses? The aggregate dynamics translate into markedly different outcomes across cohorts. Figure 6 reports the difference between the Paris Agreement and Low Carbon Price scenarios by age group (20–39, 40–59, 60–79, 80–99).

A clear age gradient emerges. Younger and middle-aged households experience large and persistent consumption losses over the transition, whereas the oldest households are largely insulated and may even record small gains in the early decades. This pattern reflects the interaction between two channels that load differently across the life cycle. The first operates through labor income. Higher carbon prices increase production costs and depress aggregate demand, reducing labor demand and compressing wages. This channel reduces household

income and consumption. The second channel operates through financial wealth. The announcement of a steeper carbon tax path generates an initial disinflationary shock, reflecting the forward-looking contraction in aggregate demand at the onset of the transition. This unexpected decline in inflation has two effects: (i) it raises the real value of nominal financial assets, generating capital gains for asset holders; and (ii) it increases the real return on financial wealth.⁹ Through this financial-wealth channel, household income and consumption rise.¹⁰ These two channels therefore operate in opposite directions. The labor-income channel reduces purchasing power, while the financial-wealth channel increases it. The overall effect on household income depends on their relative strength; other budget components are quantitatively small (Appendix C). The importance of this source-side incidence is consistent with general-equilibrium analyses showing that the distributional effects of carbon pricing depend not only on consumption patterns but also on changes in labor and capital income (Goulder et al., 2019). The interaction of these opposing channels with households' heterogeneous exposure to them defines what we call the *redistribution force* of the green transition.

Definition 1 (Redistribution force). *The redistribution force of the green transition is the reallocation of household resources generated by differences in exposure to transition-induced changes in labor income and financial wealth. In an overlapping-generations economy, these exposures vary systematically over the life cycle, giving the redistribution force an age-dependent incidence.*

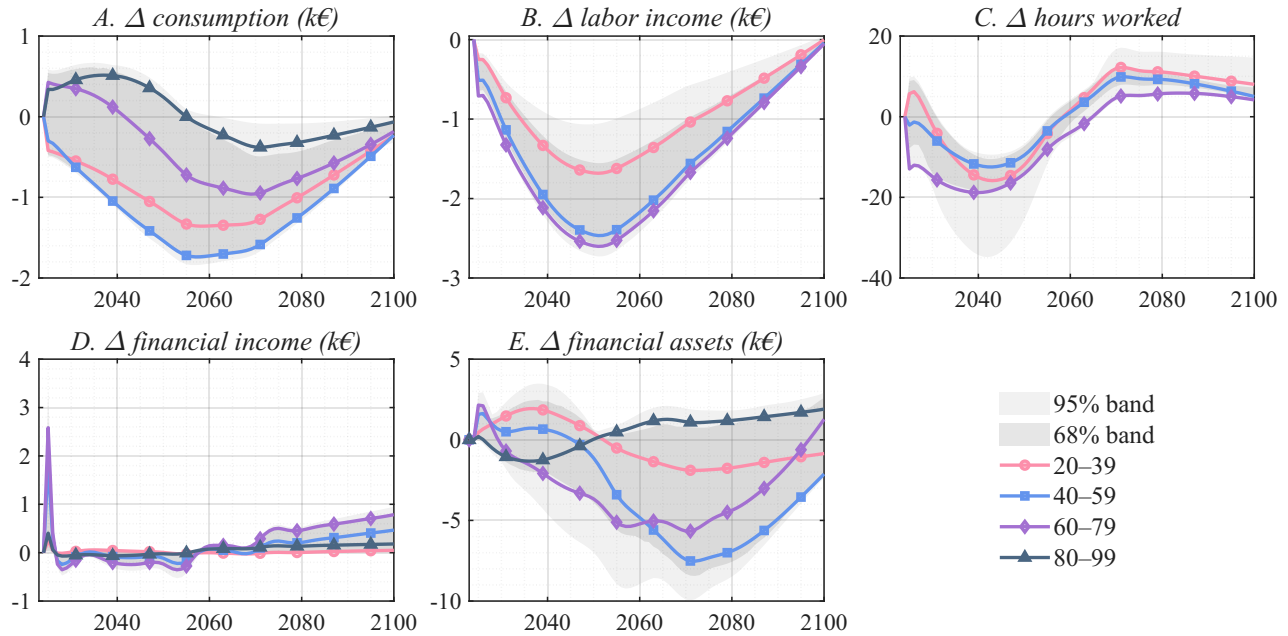
Quantitatively, the labor-income component dominates. Labor-income losses are largest for households aged 20–59, whose income is most exposed to labor-market developments, and decline sharply for older households. This age pattern is consistent with the underlying life-cycle composition of household income: labor income represents a large share of total income during working life and declines toward retirement (Appendix D). Importantly, hours worked move much less than labor income, confirming that lower real wages are the main margin of adjustment.¹¹ The financial-wealth component provides only a partial offset. All age groups experience a temporary increase in financial income at the onset of the transition, but the gains are concentrated among the 60–79 and 80–99 groups, which hold the bulk

⁹This second effect arises because the return on financial wealth depends, among other things, on the price of capital (see Equation 28), which reflects the expected return on physical capital discounted by risk-neutral investment funds. Following the initial disinflation, investment funds discount expected returns at a lower nominal rate, causing the price of capital to jump on impact and increasing the return on financial wealth.

¹⁰The financial-wealth channel is related to the capitalization channel in Karp and Rezai (2014), but its source differs: in our model, capital gains arise from the disinflationary revaluation of nominal wealth and dissipate along the transition.

¹¹This wage-intensive adjustment is consistent with empirical evidence of persistent wage declines and more transitory employment effects following carbon taxation (Wei and Yip, 2025).

FIGURE 6. Age-dependent distributional effects of the green transition



Notes: The figure reports the difference between the Paris Agreement and the Low Carbon Price scenarios for consumption, labor income, hours worked, financial income, and financial assets. Cohorts are grouped by age (20–39, 40–59, 60–79, 80–99). Shaded areas denote the cross-sectional distribution across households within each cohort: dark bands correspond to the 68% interval and light bands to the 95% interval. All real variables are expressed in efficient per capita units.

of financial assets. This effect is smaller and shorter lived than the decline in labor income. Asset dynamics subsequently diverge across age groups, reflecting anticipatory behavior. Because the carbon price follows a gradually rising, pre-announced path, households aged 20–59 anticipate that the bulk of wage compression lies ahead. While labor-income losses are initially modest, they increase saving to build a buffer against future income losses. As wage compression intensifies, however, these households reduce net saving and progressively decumulate financial assets to smooth consumption, leading to a marked deterioration in their balance sheets over the transition. Households aged 80–99, by contrast, remain largely insulated from labor-income losses and preserve, or even strengthen, their financial positions. The 60–79 group occupies an intermediate position: some households remain exposed to labor-income losses, while others have already retired and rely more heavily on financial wealth.¹²

¹²This redistribution force is remarkably insensitive to the conduct of monetary policy. As shown in Appendix E, alternative policy rules affect the timing of nominal adjustment but have little effect on the relative strength of the labor-income and financial-wealth channels or on the resulting intergenerational redistribution.

The distributional impact is reinforced by structural features of the economy. First, the short-run incidence of the carbon tax falls on labor because capital cannot adjust immediately. Capital adjusts intertemporally through investment, while labor markets clear within the period. The short-run incidence of the carbon tax therefore concentrates on wages, reinforcing the labor-income channel for working cohorts. Second, portfolio adjustment frictions limit households' ability to benefit from changes in asset returns. This illiquidity dampens the financial-wealth channel for middle-aged cohorts who might otherwise capture transitional gains through rapid portfolio reallocation, extending the quantitative dominance of the labor-income channel beyond what factor endowments alone would predict. Third, differences in marginal propensities to consume amplify the impact of income shocks, particularly for households relying on labor income. The no-borrowing constraint (5), combined with the hump-shaped income profile calibrated to the HFCS, generates a U-shaped MPC profile over the life cycle. Young households translate labor income losses nearly one-for-one into reduced consumption, while a population of wealthy hand-to-mouth households at intermediate ages, in the spirit of [Kaplan and Violante \(2014\)](#), implies that the transition remains imperfectly insurable even for cohorts with non-negligible wealth. Our analysis focuses on these source-side channels. Empirical evidence shows that carbon pricing can also redistribute through heterogeneous consumption patterns across households ([Feindt et al., 2021](#)). With a single consumption good, our model abstracts from this use-side incidence.

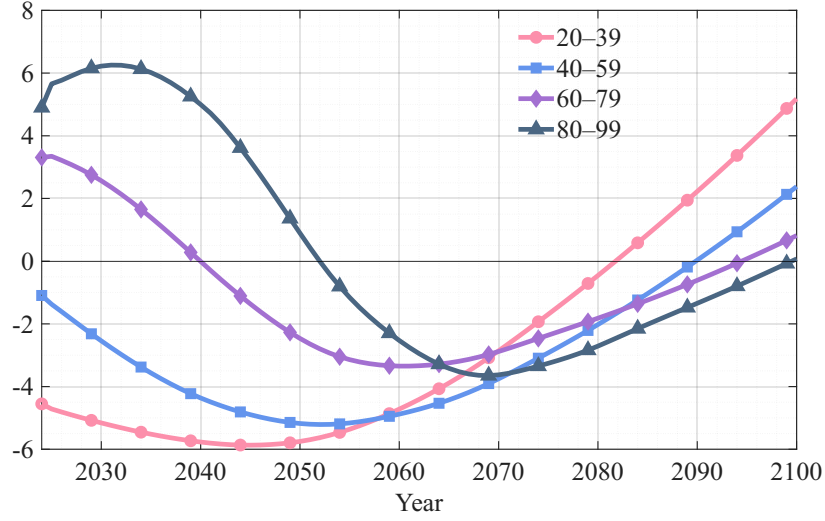
4.3. Welfare effects over the transition. The heterogeneous income and asset dynamics documented above translate into age-specific welfare effects that depend on both exposure and remaining lifetime. To quantify these effects, we compute, for each cohort, a consumption-equivalent welfare measure (CEV), defined as the proportional change in remaining lifetime consumption that makes households indifferent between the Paris Agreement and the Low Carbon Price scenarios.

Let $s \in \{\text{PA}, \text{LCP}\}$ denote the policy scenario, and let $\mathcal{W}_{j,t}^s$ denote the lifetime welfare of cohort j at time t under scenario s , as defined in Equation (1). The consumption-equivalent variation $\mathfrak{S}_{j,t}$ solves:

$$\mathcal{W}_{j,t}^{\text{PA}} = \mathbb{E}_t \left\{ \sum_{k=0}^{J-j} \beta^k \frac{N_{j+k,t+k}}{N_{j,t}} \left(\log \left((1 + \mathfrak{S}_{j,t}) c_{j+k,t+k}^{\text{LCP}} \right) - \varphi_{j+k} \frac{(h_{j+k,t+k}^{\text{LCP}})^{1+\nu}}{1+\nu} \right) \right\}. \quad (39)$$

A positive value of $\mathfrak{S}_{j,t}$ indicates that cohort j prefers the Paris Agreement, while a negative value indicates a welfare loss relative to the Low Carbon Price scenario.

FIGURE 7. Consumption-equivalent variation from the green transition



Notes: The figure reports the consumption-equivalent variation $\mathfrak{S}_{j,t}$ across cohorts (20–39, 40–59, 60–79, 80–99) along the green transition path. $\mathfrak{S}_{j,t}$ is the proportional shift in the Low Carbon Price baseline consumption stream required to match lifetime utility under the Paris Agreement. The vertical axis measures the magnitude of the consumption-equivalent variation in percent of baseline consumption.

This age gradient, however, is only a snapshot of a transition that unfolds over several decades. The forces underlying it are themselves time varying: wage compression eventually recedes, the initial financial-wealth gains dissipate, and the benefits from lower climate damages gradually materialize. Moreover, households age as these forces evolve, changing their exposure to labor income and financial wealth. Whether today’s winners remain winners, and today’s losers remain losers, is therefore an inherently dynamic question. We now trace this redistribution over time and across generations.

Figure 7 reports the resulting welfare profiles by age groups.¹³ The welfare effects exhibit a pronounced age gradient at the beginning of the transition. In 2023, households aged 20–39 experience a welfare loss of about 4.5 percent of permanent consumption, while those aged 80–99 gain about 5 percent. This ranking is the welfare counterpart of the redistribution force identified previously. Younger and middle-aged households are exposed to persistent labor-income losses over a large part of their remaining working lives and have limited financial wealth with which to absorb them. Older households, by contrast, are largely insulated

¹³Aggregating individual CEVs across age bands requires care: the discount factor $\sum_{k=0}^{J-j} \beta^k N_{j+k,t+k} / N_{j,t}$ shrinks mechanically with age and inflates the CEV of older cohorts. We address this by weighting each cohort’s CEV by both its population share and its remaining lifetime. For a band $\mathcal{G}(t)$ of cohorts alive at time t , the aggregated CEV is $\overline{\mathfrak{S}}_{\mathcal{G},t} = \sum_{j \in \mathcal{G}(t)} q_{j,t} \mathfrak{S}_{j,t}$, with $q_{j,t} = (J-j+1) N_j^{rel} / \sum_{j \in \mathcal{G}(t)} (J-j+1) N_j^{rel}$, where $J-j+1$ is the remaining lifetime of cohort j . This downweights cohorts with few remaining periods.

from wage compression and benefit more from the initial increase in the real value and return of accumulated wealth. For the youngest group, welfare losses subsequently deepen, reaching about 6 percent of permanent consumption around 2045. These losses are particularly large for households exposed to the transition during their main earning and saving years. Lower labor income reduces not only current consumption but also saving and future wealth accumulation. Retirees face the opposite configuration: they no longer bear the direct labor-income cost of the transition and enter it with larger asset positions, making them more exposed to the financial-wealth gains. The redistribution force therefore translates into a substantial reallocation of welfare across ages at the onset of the transition. The magnitude of these welfare effects exceeds that in [Fried et al. \(2024\)](#), while the policy experiments and model environments differ substantially. Their analysis considers a constant carbon tax in a real economy, whereas our framework incorporates a rising carbon price, evolving climate damages, and nominal rigidities. In particular, disinflationary asset revaluation amplifies the divergence between workers and retirees during the initial phase of the transition.

5. THE DEMOGRAPHIC FOOTPRINT OF THE GREEN TRANSITION

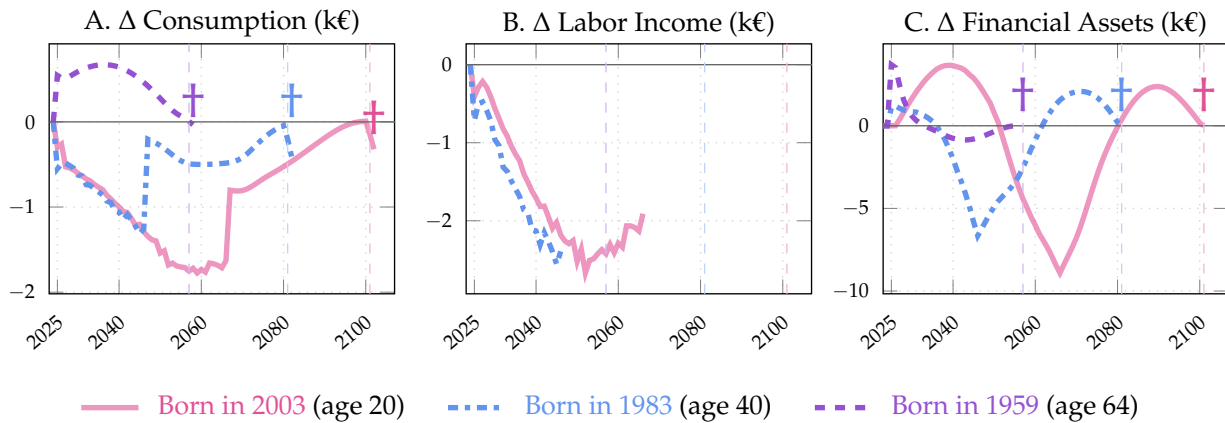
Changing incidence over time does not reveal how the transition affects a generation over its lifetime. Successive cross-sections compare different households at different stages of their life cycle and therefore conceal how gains and losses accumulate within a given generation. In particular, exposure to labor-income losses during working life can reduce saving and leave affected households with lower wealth later in life, consistent with microeconomic evidence on the persistent effects of adverse labor-market shocks on household saving and wealth ([Basten et al., 2016](#); [Barnette, 2023](#)). Along a prolonged transition, these effects can persist long after the forces that generated them have faded. This intertemporal legacy constitutes the *demographic footprint* of the green transition.

Definition 2 (Demographic footprint). *The demographic footprint of the green transition is the persistence of its distributional effects within birth cohorts as they age, beyond their direct exposure to the forces that initially generated them.*

We uncover it by following generations as they age through the transition, before comparing lifetime welfare across birth cohorts.

5.1. Following generations through the transition. Consider three illustrative birth cohorts, born in 1959, 1983, and 2003. They are respectively 64, 40, and 20 years old at the beginning of the transition in 2023 and therefore face the same policy from very different positions in their

FIGURE 8. Following generations through the green transition



Notes: The figure follows three birth cohorts through the green transition: individuals born in 1959, 1983, and 2003, who are respectively at retirement age (64), age 40, and age 20 at the onset of the transition in 2023. The panels report differences in consumption, labor income, and financial assets between the Paris Agreement and Low-Carbon Price scenarios, expressed in thousands of euros. Each line follows the same birth cohort as it ages through the transition. Vertical markers indicate the end of life of each cohort.

life cycle. The 1959 cohort is just entering retirement, the 1983 cohort is in the middle of its working life, while the 2003 cohort enters adulthood just as the transition begins. Following these three generations over their remaining lifetimes allows us to isolate how the timing of the transition relative to the life cycle shapes its incidence.

Figure 8 follows consumption, labor income, and financial assets for these three cohorts, measured as differences between the Paris-Agreement and Low-Carbon-Price scenarios. Panel A shows a striking contrast across generations. The 1959 cohort initially consumes more under the Paris-Agreement, whereas consumption falls for both working-age cohorts. The decline is particularly large and persistent for the 2003 cohort, while the 1983 cohort occupies an intermediate position. Importantly, these differences persist as the cohorts age: the consumption losses of the younger generations extend well beyond the initial phase of the transition. The labor-income channel, shown in Panel B, provides the first part of the explanation. The 1959 cohort is essentially insulated from wage compression because it has already retired when the transition begins. The 1983 cohort, by contrast, experiences substantial labor-income losses over the remainder of its working life. The 2003 cohort is exposed for even longer, as wage compression affects almost its entire working life. The timing of the transition therefore determines not only the magnitude of the labor-income loss but also the number of years over which it accumulates. The dynamics of asset holdings in Panel C provide the second part of the explanation and show how these effects propagate over the life cycle. The 1959 cohort enters the transition with substantial accumulated assets and initially

benefits from their favorable revaluation, while facing virtually no labor-income losses. The 1983 cohort combines an existing stock of wealth with continued exposure to wage compression. Lower labor income reduces saving and progressively translates into a sizeable asset shortfall before retirement. The 2003 cohort displays a more pronounced dynamic response. Entering the transition with virtually no wealth, it initially increases saving in anticipation of larger labor-income losses ahead. As the carbon price rises and wage compression intensifies, however, saving declines and the cohort progressively draws down its accumulated assets to smooth consumption, generating a substantial asset shortfall later in life.

The three cohorts show that intergenerational incidence depends on the sequence of exposures experienced over the life cycle, not simply on age at a given date. Past labor-income losses remain embedded in subsequent asset holdings, so that their effects persist after the underlying wage compression has faded. The cohort profiles thus provide a direct illustration of the demographic footprint.

5.2. The demographic footprint across generations. We now examine how the demographic footprint varies across generations. Indexing households by year of birth rather than by age at a given date, we evaluate their outcomes over their remaining lifetime. This maps the changing cross-sectional incidence documented in Section 4.3 into a single generational welfare profile.

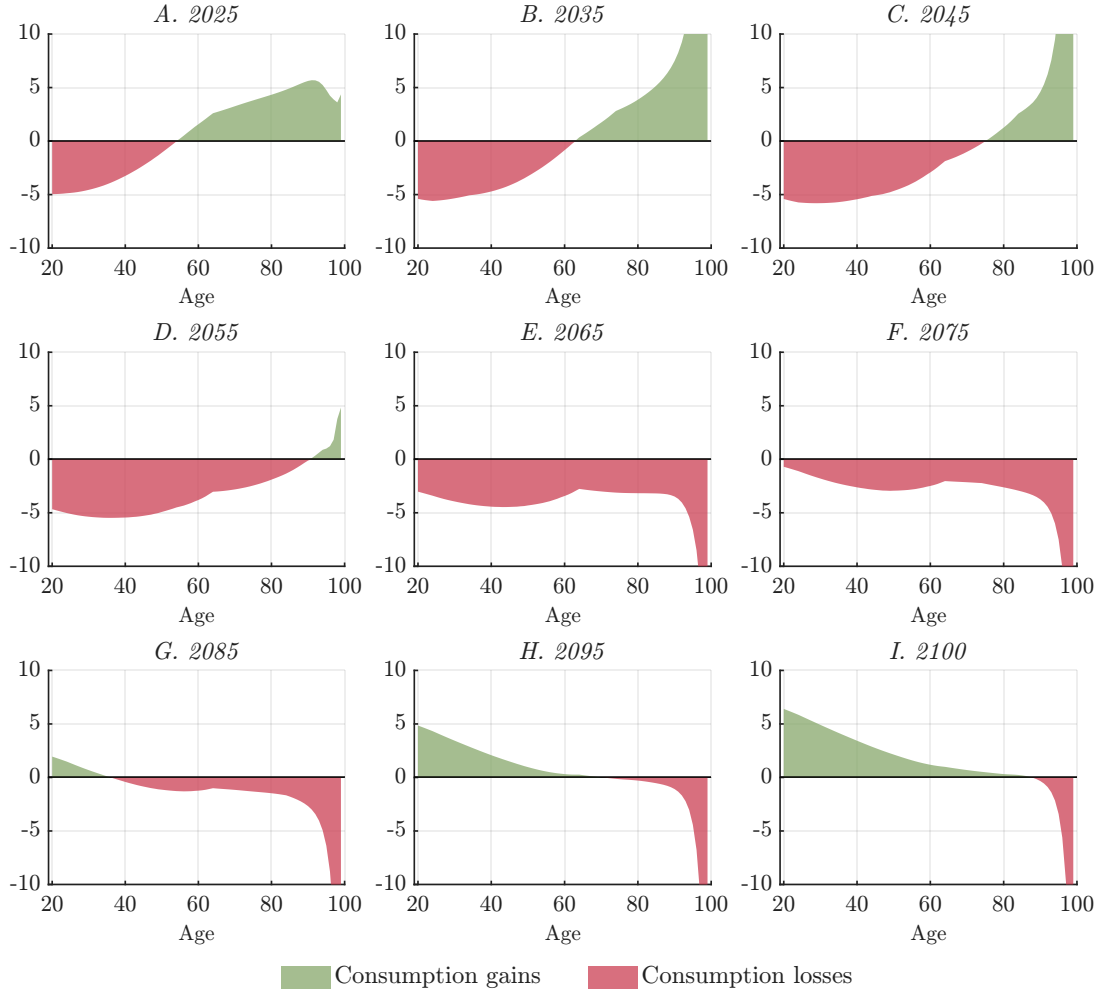
TABLE 2. Cohort-level effects of the green transition (in %)

Birth cohort	Lifetime CEV	Lifetime labor income	Lifetime financial assets
1960	2.47	0.00	0.10
1970	0.22	-3.17	-0.13
1980	-2.16	-5.10	-0.45
1990	-3.88	-6.53	-0.69
2000	-4.69	-7.38	-0.53
2010	-5.22	-7.94	-1.13
2020	-5.48	-7.24	-1.39
2030	-5.16	-5.55	-0.68
2040	-3.93	-3.49	0.51
2050	-1.93	-1.35	2.06
2060	0.60	0.92	3.65

Notes: Lifetime CEV denotes the consumption-equivalent variation, expressed as a percentage of remaining lifetime consumption, that makes a household born in the given year indifferent between the Paris Agreement and Low Carbon Price scenarios (positive values indicate a preference for the Paris Agreement). Lifetime labor income and financial assets report the difference between the mean levels under the two scenarios, divided by the absolute mean Low Carbon Price level. Lifetime financial assets refer to the asset stock rather than the income earned on it.

Table 2 reports consumption-equivalent welfare changes for birth cohorts from 1960 to 2060, together with their lifetime changes in labor income and financial assets. The resulting welfare profile is strongly non-monotonic. The 1960 cohort gains substantially from the transition, with a lifetime CEV of 2.47 percent. Welfare then declines rapidly across successive cohorts: the CEV is close to zero for the 1970 cohort, turns negative for the 1980 cohort, and reaches its lowest level for cohorts born around 2010–2020, with losses of about 5.5 percent of remaining lifetime consumption. For later-born cohorts, the pattern reverses. Welfare losses progressively shrink and eventually turn into a gain for the 2060 cohort. Welfare losses are

FIGURE 9. Age profile of welfare costs along the transition path



Notes: Each panel reports the consumption-equivalent variation $\mathfrak{S}_{j,t}$ across cohorts aged 20–99 at several dates along the transition path. $\mathfrak{S}_{j,t}$ is the proportional shift in the Low Carbon Price baseline consumption stream required to match lifetime utility under the Paris Agreement. The vertical axis measures the magnitude of the consumption-equivalent variation in percent of baseline consumption, where red (resp. green) shading indicates a welfare loss (resp. gain), expressed in consumption-equivalent terms, under the Paris Agreement relative to Low Carbon Price. The horizontal axis reflects the age distribution of living cohorts at each date.

thus concentrated among generations whose working lives overlap most strongly with the costly phase of the transition. Generations already close to retirement when carbon prices rise avoid most of the wage compression and benefit from their accumulated wealth. Generations entering the labor market around the transition are exposed to its costs over much of their working lives. Later generations, in contrast, face progressively smaller transition costs and enjoy a larger share of the long-run benefits from climate stabilization. The lifetime income and asset measures clarify why welfare cannot be inferred from labor-income exposure alone. Labor-income losses and financial assets display distinct profiles across cohorts, as wage compression affects both current resources and subsequent wealth accumulation. Consequently, the cohort experiencing the largest labor-income loss need not be the cohort experiencing the largest welfare loss. These generational welfare differences reflect the persistence underlying the demographic footprint. Generations born around the turn of the century bear particularly large welfare costs because their exposure to the costly phase of the transition affects not only contemporaneous income but also the wealth they carry into subsequent stages of life. The cohort analysis thus shows how successive exposures accumulate into persistent differences in lifetime welfare.

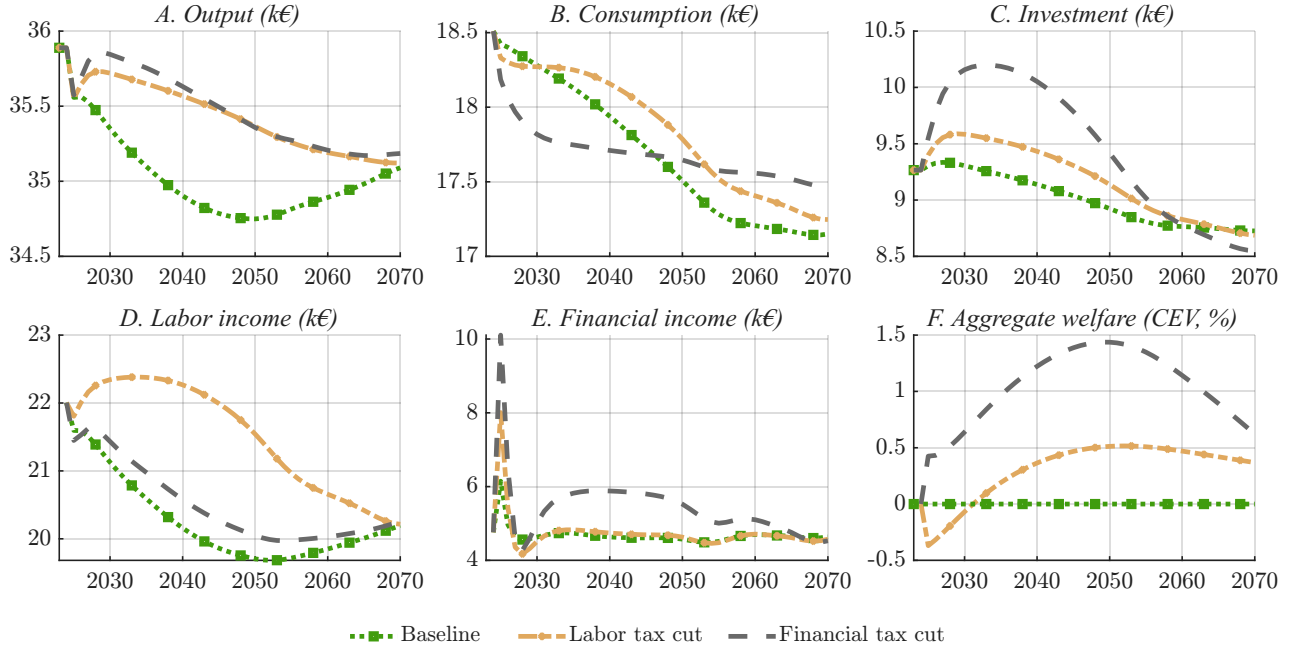
Figure 9 brings together the age and time dimensions of incidence by displaying the full age profile of welfare effects at selected dates along the transition. The boundary between winners and losers shifts markedly over time: gains are initially concentrated among older households, while losses fall primarily on working-age households; later in the transition, this pattern progressively reverses. The figure therefore highlights why age and generation provide distinct but complementary views of incidence: the welfare effect associated with a given age depends on when that age is reached along the transition path.

6. CARBON REVENUE RECYCLING: BEYOND THE DOUBLE DIVIDEND

Carbon pricing generates an environmental dividend, while recycling its revenues through reductions in distortionary taxes can generate a second dividend through higher aggregate welfare. In our intergenerational setting, recycling can also generate a *distributional third dividend* by improving intergenerational compensation while reducing consumption inequality. We identify a policy range over which these three dividends can be jointly achieved.

6.1. The double dividend. A standard argument for recycling carbon-tax revenues through reductions in distortionary taxes is the possibility of a double dividend: environmental benefits are accompanied by welfare gains from reducing pre-existing tax distortions (Goulder,

FIGURE 10. Aggregate effects of alternative carbon revenue recycling schemes



Notes: The figure displays transition paths under the Paris-Agreement scenario for three carbon revenue recycling schemes: baseline (lump-sum transfers to households), labor income tax cut, and financial income tax cut. Panel F reports the aggregate consumption-equivalent variation (CEV) of each tax-cut scheme relative to the Paris-Agreement baseline with lump-sum transfers.

1995, Bovenberg and Heijdra, 1998). We examine this possibility by considering two alternative recycling schemes that target the two main sources of household income. In the first, a fraction $\zeta_L \in [0, 1]$ of carbon-tax revenues is used to reduce labor-income taxation:

$$(\tau_h - \tau_{h,t})W_t h_t = \zeta_L \tau_{e,t} e_t, \quad (40)$$

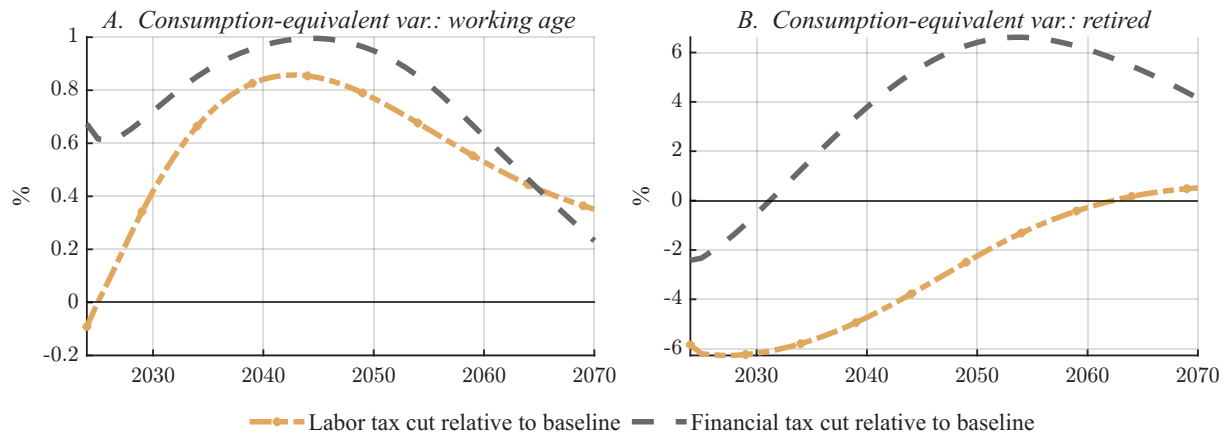
so that $\zeta_L = 0$ corresponds to the baseline lump-sum scheme and $\zeta_L = 1$ implements a full labor double dividend. In the second, a fraction $\zeta_F \in [0, 1]$ is instead used to reduce financial-income taxation:

$$(\tau_a - \tau_{a,t})(R_t^a - 1)P_{t-1}a_t = \zeta_F \tau_{e,t} e_t. \quad (41)$$

In both cases, the government budget remains balanced period by period, so that the two experiments differ only in the margin through which carbon revenues are recycled.

Figure 10 reports the aggregate effects of the two recycling schemes relative to lump-sum redistribution. Both schemes improve aggregate outcomes, although through different margins. Reducing labor-income taxation primarily supports consumption and output by cushioning the decline in disposable labor income, whereas reducing financial-income taxation stimulates capital accumulation more strongly, generating larger gains in investment and output (Barrage, 2020). These macroeconomic gains also translate into higher aggregate

FIGURE 11. Intergenerational welfare effects of carbon revenue recycling



Notes: The panels show the welfare gap between a given revenue recycling scheme and the baseline. Results are reported separately for the working-age population (aged 20–64) and the retired (aged 65–99). The baseline corresponds to the Paris-Agreement scenario, in which carbon revenue is redistributed to households through lump sum transfers.

welfare when evaluated over the transition, measured in consumption-equivalent variation (CEV), consistent with the conventional double-dividend mechanism. The gain is larger under financial-income tax relief, reflecting its stronger effect on capital accumulation and aggregate activity. This stronger role of financial-income tax relief echoes [Fried et al. \(2024\)](#), who find substantial welfare gains from capital-income tax recycling in the United States.

Yet these aggregate welfare gains conceal substantial differences across generations. As shown above, the composition of household income changes systematically over the life cycle, so the tax base through which carbon revenues are recycled determines who benefits from the second dividend (Appendix F). Figure 11 shows that working-age households benefit from both recycling schemes over most of the transition, whereas the welfare effects for retirees differ sharply across instruments. Labor-income tax relief lowers retiree welfare relative to lump-sum recycling over much of the transition, while financial-income tax relief generates substantial welfare gains. These contrasting welfare effects reflect the underlying income channels: labor-income tax relief primarily benefits working-age households, whereas financial-income tax relief disproportionately benefits older, asset-rich households. The aggregate second dividend is therefore not distributionally neutral: its incidence across age groups depends on how carbon revenues are recycled.

6.2. A distributional third dividend. The aggregate second dividend documented above does not fully characterize the distributional consequences of revenue recycling. We therefore ask whether recycling can generate a *distributional third dividend*. This dividend has two dimensions. Intergenerational compensation captures whether the welfare gains from recycling are shared across age groups, while consumption inequality captures how these gains

are distributed across households. We measure the latter using the cross-sectional Gini coefficient of household consumption, computed across all households alive at each date.

Rather than relying on either polar scheme, we therefore consider a mixed recycling strategy that combines labor- and financial-income tax relief. Let $\zeta \in [0, 1]$ denote the share of carbon revenues allocated to financial-income tax cuts, with the remaining share $1 - \zeta$ used to reduce labor-income taxation. The corresponding tax adjustments satisfy

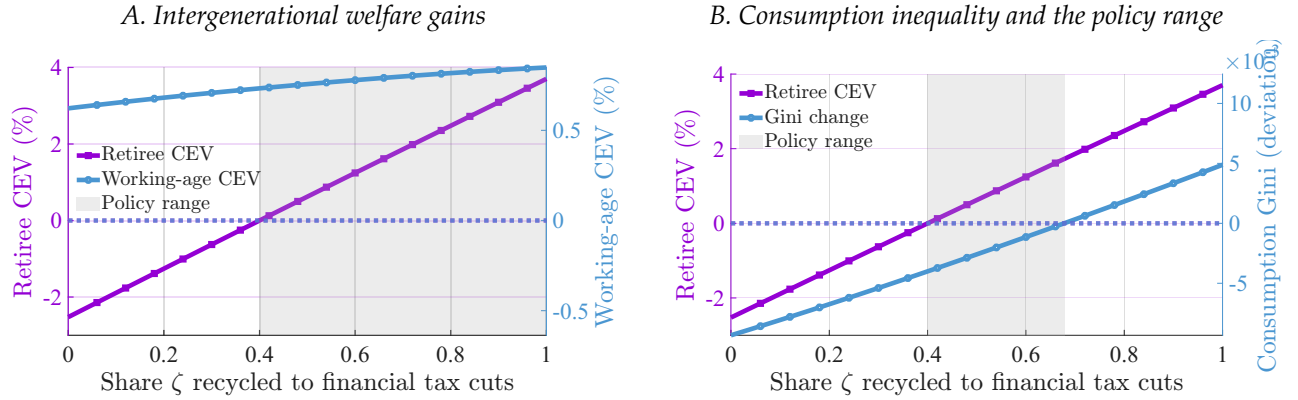
$$(\tau_a - \tau_{a,t})(R_t^a - 1)P_{t-1}a_t = \zeta\tau_{e,t}e_t, \quad (42)$$

$$(\tau_h - \tau_{h,t})W_th_t = (1 - \zeta)\tau_{e,t}e_t. \quad (43)$$

Thus, $\zeta = 0$ corresponds to full recycling through labor-income tax relief, whereas $\zeta = 1$ corresponds to full recycling through financial-income tax relief. Varying ζ between these two polar cases allows us to characterize whether the aggregate second dividend can be accompanied by improvements along both distributional dimensions.

We evaluate these recycling mixes against the two distributional dimensions of the third dividend. Figure 12 shows how both vary with the composition of revenue recycling. Both are evaluated cross-sectionally at each date and relative to the lump-sum baseline, and then averaged over the full transition horizon. Welfare is measured by the mean consumption-equivalent variation separately for working-age and retired households, while consumption inequality is measured by the Gini coefficient across all living households. Each measure is then averaged over the full transition horizon, yielding one summary statistic for each value of ζ . We first require recycling to generate welfare gains for both age groups. Panel A shows that working-age households gain relative to lump-sum recycling throughout the entire range of ζ , whereas retiree welfare increases with the share of revenues allocated to financial-income tax relief and becomes positive at $\zeta \simeq 0.40$. The welfare requirement therefore imposes a lower bound on ζ , defining an intergenerationally welfare-improving range of $\zeta \in [0.40, 1]$. We then impose the additional requirement that these welfare gains not come at the cost of higher consumption inequality. Panel B shows that, as ζ increases, the inequality-reducing effect of recycling progressively weakens, with the consumption Gini reaching its baseline level at $\zeta \simeq 0.68$. The inequality criterion therefore imposes an upper bound on ζ and further restricts the welfare-improving range. Combining the two requirements yields the policy range $\zeta \in [0.40, 0.68]$. These two conditions define the distributional third dividend.

FIGURE 12. Carbon revenue recycling: The third dividend range



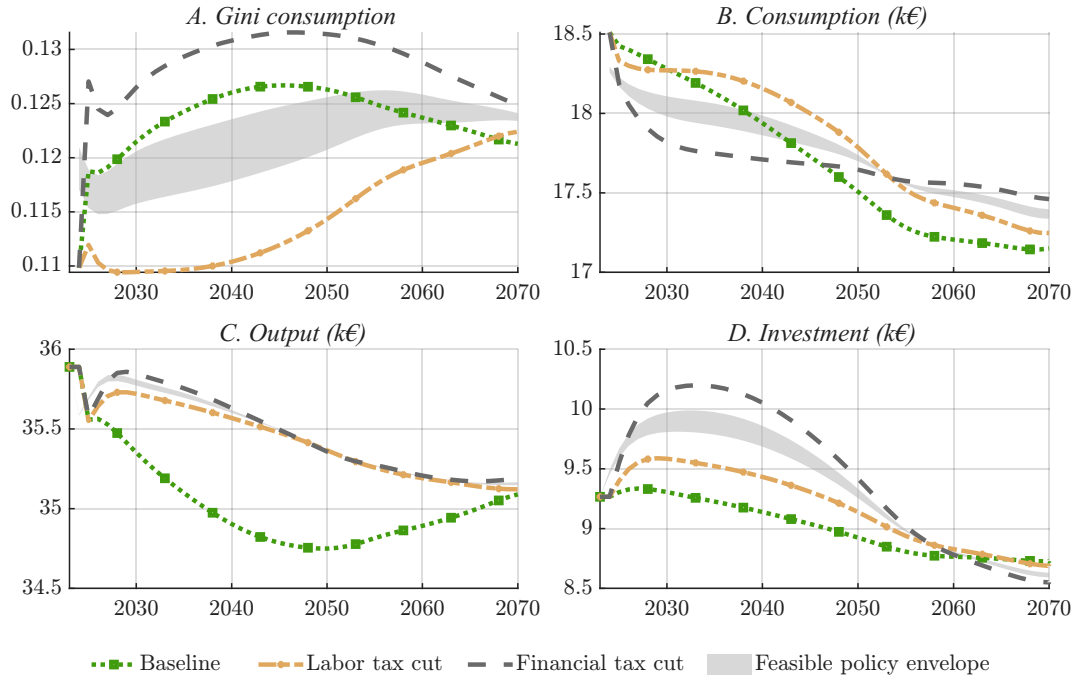
Notes: Panel A reports the average consumption-equivalent variation (CEV) of working-age and retired households relative to lump-sum recycling as a function of the share ζ of carbon revenues recycled through financial-income tax cuts. The shaded area, $\zeta \in [0.40, 1]$, identifies the range over which both groups experience welfare gains. Panel B adds the change in the consumption Gini relative to the lump-sum baseline. Requiring consumption inequality not to increase narrows the policy range to $\zeta \in [0.40, 0.68]$. All measures are averaged over the transition horizon.

Definition 3 (Distributional third dividend). *A revenue-recycling scheme delivers a distributional third dividend when, in addition to the aggregate second dividend, its welfare gains are shared across working-age and retired households and consumption inequality is reduced relative to lump-sum recycling.*

Figure 13 illustrates the transition dynamics generated by the policy range over the 2025–2070 horizon. The shaded area reports the paths associated with $\zeta \in [0.40, 0.68]$. Relative to lump-sum recycling, these schemes generally support higher output and investment during the costly phase of the transition, reflecting the partial allocation of carbon revenues to financial-income tax relief. Consumption initially remains below the baseline, until around 2045, as satisfying the distributional requirements limits the short-run support provided by the financial-wealth channel. As capital accumulates, however, consumption recovers and eventually rises above the baseline. The policy-range envelope lies between the two polar recycling schemes: relative to labor-income tax relief, it provides stronger support to capital accumulation and output, while relative to financial-income tax relief, it limits the increase in consumption inequality. The policy range therefore preserves the aggregate second dividend while delivering the distributional third dividend through intergenerational compensation and lower consumption inequality.

We assess the robustness of the policy range in Appendix G. We consider six alternative specifications covering nominal rigidities, the carbon-price path, demographic structure, and the retirement age. In each case, we recompute the policy range using the same two distributional requirements. The range remains non-empty across all specifications, although its

FIGURE 13. The third dividend: Transition dynamics within the policy range



Notes: The figure displays transition paths under the Paris-Agreement scenario for three carbon revenue recycling schemes: baseline (lump-sum transfers), labor-income tax cut, and financial-income tax cut, together with the policy-range envelope. The envelope (grey area) spans the paths generated by recycling shares $\zeta \in [0.40, 0.68]$, for which both working-age and retired households experience welfare gains relative to the baseline and consumption inequality does not increase.

boundaries shift quantitatively. The third dividend is therefore not specific to the baseline calibration: some financial-income tax relief is required to achieve intergenerational compensation, while excessive reliance on it prevents a reduction in consumption inequality.

6.3. The limits of redistribution. The distributional third dividend does not imply that every birth cohort benefits from revenue recycling. The requirements defining the policy range ensure that both working-age and retired households gain relative to lump-sum recycling and that consumption inequality is reduced, but they do not require welfare improvements for every birth cohort over its remaining lifetime. This distinction matters because the two distributional dimensions are evaluated cross-sectionally and then averaged over the transition, whereas lifetime welfare reflects the entire sequence of exposures experienced by a generation as it ages. A policy that improves welfare across age groups and reduces inequality on average over the transition may therefore leave the cumulative incidence across birth cohorts largely unchanged, or even reinforce it for some generations.

Table 3 reports lifetime consumption-equivalent welfare changes across birth cohorts under the baseline Paris Agreement scenario, the two polar recycling schemes, and the policy range.

TABLE 3. Lifetime welfare effects across carbon revenue recycling schemes (in %)

Birth cohort	Baseline	Labor-income subsidy	Financial-income subsidy	Policy range
1960	2.47	-1.52	3.44	[0.58, 1.67]
1970	0.22	-2.09	2.30	[-0.19, 0.77]
1980	-2.16	-2.21	-0.28	[-1.34, -0.92]
1990	-3.88	-2.65	-3.26	[-3.00, -2.86]
2000	-4.69	-3.33	-5.34	[-4.59, -4.14]
2010	-5.22	-4.28	-6.81	[-5.88, -5.33]
2020	-5.48	-5.07	-6.93	[-6.24, -5.83]
2030	-5.16	-4.99	-5.68	[-5.42, -5.27]
2040	-3.93	-3.74	-3.80	[-3.78, -3.76]
2050	-1.93	-1.62	-1.74	[-1.70, -1.67]
2060	0.60	0.81	0.54	[0.64, 0.70]

Notes: The table reports lifetime consumption-equivalent variations (CEV) by birth cohort under alternative carbon revenue recycling schemes. The baseline corresponds to the Paris-Agreement scenario, in which carbon revenues are redistributed through lump-sum transfers. The alternative schemes recycle revenues through labor-income tax cuts, financial-income tax cuts, or mixed recycling within the policy range, $\zeta \in [0.40, 0.68]$. Positive (negative) CEV values indicate welfare gains (losses) relative to the Low-Carbon-Price scenario.

It reveals substantial heterogeneity across generations. The 1960 cohort remains a winner throughout the policy range, with lifetime welfare gains between 0.58 and 1.67 percent of permanent consumption. For the 1970 cohort, the effect ranges from -0.19 to 0.77 percent, so whether it is fully compensated depends on the precise recycling mix. From the 1980 cohort onward, however, lifetime welfare effects remain negative throughout the policy range, and it is only for the generation born in 2060 that they turn positive again, with gains ranging from 0.64 to 0.70 percent. The third dividend can therefore improve the distribution of welfare across age groups without eliminating the demographic footprint across birth cohorts.

Most strikingly, recycling within the policy range deepens the lifetime welfare losses of generations born between 2010 and 2030 relative to lump-sum recycling. The 2010 cohort loses 5.22 percent of permanent consumption under the baseline, compared with between 5.33 and 5.88 percent within the policy range. For the 2020 cohort, which experiences the largest lifetime loss under the baseline, the loss increases from 5.48 percent to between 5.83 and 6.24 percent. The same pattern holds for the 2030 cohort, whose loss rises from 5.16 percent to between 5.27 and 5.42 percent. For these generations, therefore, delivering the third dividend does not merely fail to provide lifetime compensation; it increases the cumulative welfare cost of the transition.

This pattern reflects the distinction between improving the cross-sectional distribution of welfare and redistributing welfare across generations over their lifetimes. The policy range

is constructed to ensure welfare gains for both working-age and retired households while reducing consumption inequality, but these criteria do not account for when particular birth cohorts are exposed to the underlying recycling instruments. Cohorts born between 2010 and 2030 spend a substantial part of their working lives during the costly phase of the transition and remain strongly exposed to labor-income losses. Satisfying the retiree-welfare requirement diverts part of carbon revenues from labor-income tax relief toward financial-income tax relief, reducing the support received by these cohorts during their working lives. Revenue recycling can therefore improve the targeted distributional criteria while shifting part of the cumulative adjustment burden toward generations whose working lives overlap most strongly with the costly phase of the transition.

This limitation has a direct policy implication. If lifetime intergenerational equity is an explicit objective, adjusting the composition of carbon-revenue recycling alone may be insufficient. Additional instruments targeted at the cohorts bearing the largest lifetime transition costs may be needed. More generally, compensation policies should account not only for the distribution of gains and losses at a given date, but also for how these effects accumulate over the life cycle. Who ultimately pays therefore depends not only on how carbon revenues are redistributed, but also on when transition costs intersect each generation's life cycle.

7. CONCLUSION

This paper studies the dynamic intergenerational incidence of the green transition in an overlapping-generations macro-climate framework. Carbon pricing generates a *redistribution force* through two opposing channels: a labor-income channel, driven by wage compression during the abatement phase, and a financial-wealth channel, operating through asset revaluation and financial returns. Because households' exposure to these channels changes over the life cycle while their relative strength evolves along the transition, who bears its costs depends critically on when we look.

Our analysis delivers three main findings. First, the incidence of carbon pricing changes along the transition. Working-age households initially bear most of the costs through lower labor income, while retirees benefit from their greater exposure to financial wealth. As the transition unfolds, however, the relative strength of these channels changes and so does the identity of winners and losers. Second, following birth cohorts over their lifetimes reveals a persistent *demographic footprint*. Generations exposed to wage compression during a large part of their working lives consume and save less and enter retirement with lower accumulated wealth. Distributional effects can therefore persist long after the macroeconomic forces

that generated them have faded. Third, carbon-revenue recycling can extend the conventional double dividend to a *distributional third dividend*. Allocating 40 to 68 percent of revenues to financial-income tax relief allows welfare gains to be shared across age groups while reducing consumption inequality. Yet this third dividend does not eliminate the demographic footprint and can deepen lifetime welfare losses for some generations.

The broader implication is that the incidence of a long-lasting policy transition cannot be assessed from a snapshot of winners and losers. When relative incomes change over decades, households age while the economic effects of the policy evolve, and income losses experienced during working life can have persistent distributional consequences through lower wealth accumulation. Climate-policy design should therefore account not only for who gains and loses at each date, but also for how these gains and losses accumulate over generations' lifetimes. Revenue recycling can substantially reshape this incidence, but intergenerational neutrality may require additional instruments targeted at the generations bearing the largest lifetime transition costs.

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ONLINE APPENDIX
–NOT INTENDED FOR PUBLICATION–

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APPENDIX A. EQUILIBRIUM CONDITIONS OF THE MODEL

This appendix presents the equilibrium conditions of the model.

All the aggregates are now expressed in efficiency units, and we use a tilde to denote such variables. The relative carbon price is defined as $\bar{\tau}_{e,t} = \tau_{e,t}/P_t$, the real rental rate of capital is given by $r_t^k = R_t^k/P_t$, while the price of a unit of capital expressed in real terms is $q_t = Q_t/P_t$. We define mc_t as the real marginal cost of production for intermediate-goods producers, gross of environment-related costs (abatement and policy), while the term $\omega_t > 0$ denotes the component of the marginal cost associated with the manufacturing process. We define the Lagrange multiplier on the no-borrowing constraint (the Kuhn-Tucker multiplier) of households as $\tilde{\lambda}_{j,t}^a$. Finally, we denote by g public spending as a share of output.

Demographics.

- $N_t = \sum_{j=1}^J N_{j,t}$
- $N_t = (1+n)N_{t-1}$
- $N_j^{rel} = N_{j,t}/N_t$, for $j \in [1 : J]$
- $N_j^{rel} = \frac{1-\omega_{j-1}}{1+n} N_{j-1}^{rel}$, for $j \in [2 : J]$

Household sector.

- $(1+g_{z,t+1})\tilde{a}_{j+1,t+1} = \tilde{x}_{j,t} + \frac{1}{\pi_t}[1+(1-\tau_{a,t})(R_t^a-1)]\tilde{a}_{j,t} - \frac{\kappa_x}{2}\tilde{x}_{j,t}^2$,
for $j \in [1 : (J-1)]$

- $\tilde{x}_{j,t} = (1 - \tau_{h,t})\tilde{w}_t z_j h_{j,t} + \widetilde{beq}_t + \tilde{\tau}_{j,t} - \tilde{c}_{j,t}$
for $j \in [1 : (JR - 1)]$
- $\tilde{x}_{j,t} = \widetilde{beq}_t + \tilde{\tau}_{j,t} - \tilde{c}_{j,t}$
for $j \in [JR : J]$
- $\tilde{a}_{1,t} = 0$
- $\tilde{a}_{J+1,t+1} = 0$
- $\varphi_j h_{j,t}^\nu = \frac{z_{j,t} \tilde{w}_t (1 - \tau_{h,t})}{\tilde{c}_{j,t}}$, for $j \in [1 : (JR - 1)]$
- $h_{j,t} = 0$, for $j \in [JR : J]$
- $1 = \beta(1 - \omega_j) \mathbb{E}_t \left\{ \frac{\tilde{c}_{j,t}}{(1 + g_{z,t+1}) \tilde{c}_{j+1,t+1}} \frac{1 + (1 - \tau_{a,t+1})(R_{t+1}^a - 1)}{\pi_{t+1}} \frac{1 - \kappa_x \tilde{x}_{j,t}}{1 - \kappa_x \tilde{x}_{j+1,t+1}} \right\} + \tilde{\lambda}_{j,t}^a$, for $j \in [1 : (J - 1)]$
- $\tilde{a}_{j+1,t+1} \geq 0$, $\tilde{\lambda}_{j,t}^a \geq 0$ and $\tilde{\lambda}_{j,t}^a \tilde{a}_{j+1,t+1} = 0$, for $j \in [1 : (J - 1)]$

Aggregation.

- $\tilde{c}_t = \sum_{j=1}^J N_j^{rel} \tilde{c}_{j,t}$
- $h_t = \sum_{j=1}^{JR-1} N_j^{rel} z_j h_{j,t}$
- $\tilde{a}_{t+1} = \sum_{j=1}^{J-1} \frac{N_j^{rel}}{1+n} \tilde{a}_{j+1,t+1}$
- $\widetilde{beq}_t = \sum_{j=2}^J \frac{\omega_{j-1} N_{j-1}^{rel}}{1+n} \frac{[1 + (1 - \tau_{a,t})(R_t^a - 1)] \tilde{a}_{j,t}}{\pi_t}$
- $\tilde{\tau}_t = \sum_{j=1}^J N_j^{rel} \tilde{\tau}_{j,t}$

Financial sector.

- $(1 + n)(1 + g_{z,t+1}) \tilde{k}_{t+1} = (1 - \delta) \tilde{k}_t + \left[1 - \frac{\kappa_i}{2} \left(\frac{\tilde{i}_t}{\tilde{i}_{t-1}} - 1 \right)^2 \right] \tilde{i}_t$
- $1 = q_t \left[1 - \frac{\kappa_i}{2} \left(\frac{\tilde{i}_t}{\tilde{i}_{t-1}} - 1 \right)^2 - \kappa_i \left(\frac{\tilde{i}_t}{\tilde{i}_{t-1}} - 1 \right) \frac{\tilde{i}_t}{\tilde{i}_{t-1}} \right] +$
 $+ \mathbb{E}_t \left\{ q_{t+1} \frac{\pi_{t+1}}{R_t} \kappa_i \left(\frac{\tilde{i}_{t+1}}{\tilde{i}_t} - 1 \right) \left(\frac{\tilde{i}_{t+1}}{\tilde{i}_t} \right)^2 (1 + n)(1 + g_{z,t+1}) \right\}$
- $\mathbb{E}_t \left\{ \pi_{t+1} [u_{t+1} r_{t+1}^k - \mathcal{C}_{t+1}^u / k_{t+1} + (1 - \delta) q_{t+1}] \right\} = R_t q_t$
- $r_t^k = r^k \exp \left(\frac{\psi_k}{1 - \psi_k} (u_t - 1) \right)$
- $\tilde{a}_{t+1} - \tilde{b}_{t+1} = [q_t (1 - \delta) \tilde{k}_t + \tilde{i}_t] / [(1 + n)(1 + g_{z,t+1})]$
- $\frac{R_t^a}{\pi_t} \tilde{a}_t - \frac{R_{t-1}^a}{\pi_t} \tilde{b}_t = [u_t r_t^k + (1 - \delta) q_t] \tilde{k}_t + \tilde{a}_t - \frac{1 - \psi_k}{\psi_k} r^k \left[\exp \left(\frac{\psi_k}{1 - \psi_k} (u_t - 1) \right) - 1 \right] \tilde{k}_t$

Business sector.

- $\tilde{y}_t = (1 - \mathcal{D}(T_t)) (u_t \tilde{k}_t)^\alpha h_t^{1-\alpha}$
- $mc_t = \sigma_t \tilde{\tau}_{e,t} (1 - \mu_t) + \phi_{1,t} \mu_t^{\phi_2} + \omega_t$

- $\tilde{w}_t = (1 - \alpha) \omega_t \frac{\tilde{y}_t}{h_t}$
- $r_t^k = \alpha \omega_t \frac{\tilde{y}_t}{u_t \tilde{k}_t}$
- $\tilde{d}_t = \left(1 - mc_t - \frac{\kappa_p}{2} (\pi_t - \pi)^2\right) \tilde{y}_t$
- $\tilde{\tau}_{e,t} = \frac{\phi_{1,t} \phi_{2,t} \mu_t^{\phi_2 - 1}}{\sigma_t}$
- $\tilde{e}_t = \sigma_t (1 - \mu_t) \tilde{y}_t$
- $1 - \frac{\varepsilon_p}{\varepsilon_p - 1} + mc_t \frac{\varepsilon_p}{\varepsilon_p - 1} - \kappa_p (\pi_t - \pi) \pi_t + \frac{\kappa_p}{R_t} \mathbb{E}_t \left\{ (\pi_{t+1} - \pi) \pi_{t+1}^2 \frac{\tilde{y}_{t+1} (1 + g_{z,t+1})}{\tilde{y}_t} \right\} = 0$

Public sector.

- $\tilde{g}_t = g \tilde{y}_t$
- $(1 + n) (1 + g_{z,t+1}) \tilde{b}_{t+1}^g = \frac{R_{t+1}}{\pi_t} \tilde{b}_t^g + \tilde{g}_t - \tilde{\tau}_{e,t} \tilde{e}_t + \tilde{\tau}_t - [\tau_{a,t} (R_t^a - 1)] \frac{\tilde{a}_t}{\pi_t} - \tau_{h,t} \tilde{w}_t h_t$
- $\frac{R_t}{R} = \left(\frac{R_{t-1}}{R}\right)^{\varphi_r} \left[\left(\frac{\pi_t}{\pi}\right)^{\varphi_\pi} \left(\frac{\tilde{y}_t}{\tilde{y}_{t-1}}\right)^{\varphi_y}\right]^{(1-\varphi_r)}$

Market clearing and pollution.

- $\tilde{b}_t = \tilde{b}_t^g$
- $\tilde{y}_t = \tilde{c}_t + \tilde{i}_t + \tilde{g}_t + \frac{\kappa_p}{2} (\pi_t - \pi)^2 \tilde{y}_t + \phi_{1,t} \mu_t^{\phi_2} \tilde{y}_t + \frac{\kappa_x}{2} \sum_{j=1}^J N_j^{rel} \tilde{x}_{j,t}^2 + \frac{1 - \psi_k}{\psi_k} r^k \left[\exp \left(\frac{\psi_k}{1 - \psi_k} (u_t - 1) \right) - 1 \right] \tilde{k}_t$
- $\mathcal{D}(T_t) = \gamma_1 T_t + \gamma_2 T_t^2$
- $T_t = T_{t-1} + \xi_T \bar{E}_t^{tot}$
- $E_t = Z_t N_t \tilde{e}_t = \sigma_t (1 - \mu_t) Z_t N_t \tilde{y}_t$

Trends.

- $Z_t = (1 + g_{z,t}) Z_{t-1}$
- $g_{z,t} = g_{z,t-1} (1 - \delta_z)$
- $\phi_{1,t} = \phi_{1,t-1} (1 - \delta_\phi)$
- $\sigma_t = \sigma_{t-1} (1 - \delta_\sigma)$

Rest of the world.

- $E_t^* = E_t^{tot} - E_t$

APPENDIX B. HFCS AGGREGATION PROCEDURE

This appendix describes the step-by-step aggregation procedure applied to each wave of the Household Finance and Consumption Survey.

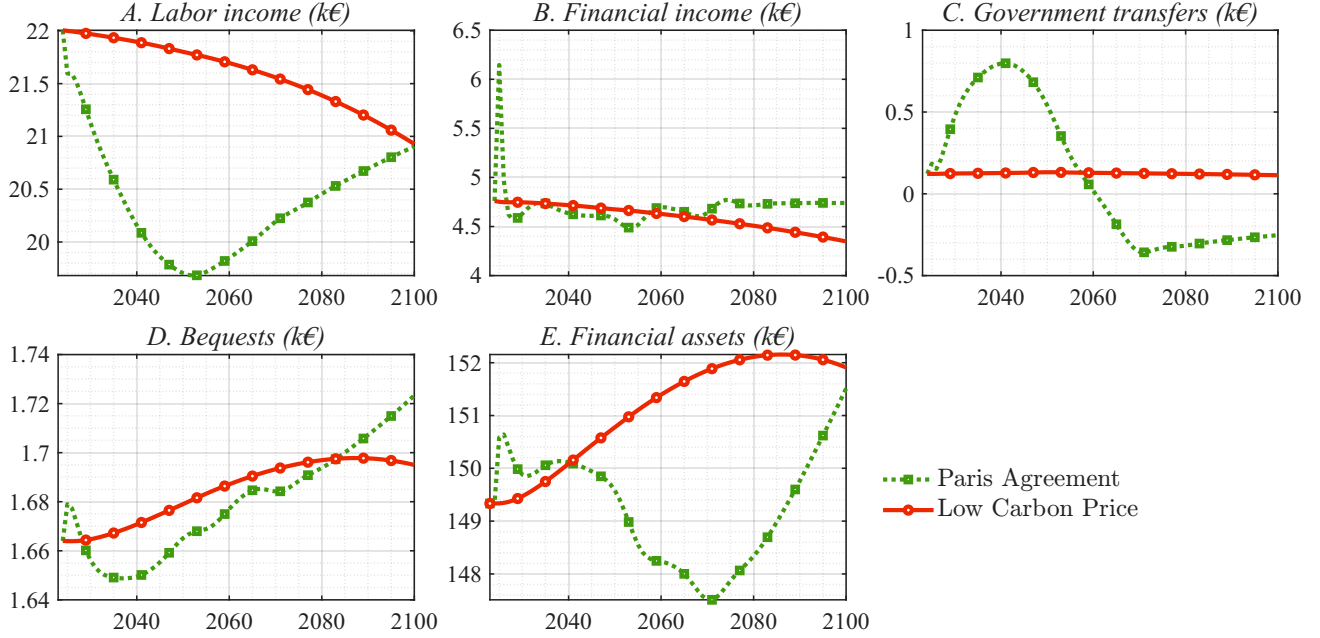
In each wave, we aggregate HFCS microdata by age in six steps. First, missing values in financial variables are set to zero so that all households enter the sample regardless of asset or income category. Second, all variables are divided by the square root of household size (DH0001) to obtain individual-equivalent quantities. Third, we exploit the multiple-imputation structure of the HFCS: for each combination of age (DHAGEH1), country (SA0100), and implicate (IM0100), we compute weighted means using the household survey weights, then average across the five implicates to obtain point estimates. Fourth, we construct the variables of interest described in the main text. Fifth, monetary amounts are deflated using the harmonized index of consumer prices at the country level (see Table B.1), expressing all quantities in constant 2021 euros. Sixth, we renormalize age-group weights to match the actual age distribution in the euro area, computed from the HFCS age variable (RA0300), so that the final euro-area aggregates reflect the true demographic composition rather than the sample composition.

TABLE B.1. Inflation adjustment factors (*relative to 2021 euros*)

	2010	2014	2017	2021
Belgium	1.2131	1.1241	1.0738	1.0000
Germany	1.1780	1.0997	1.0695	1.0000
Estonia		1.1534	1.0980	1.0000
Ireland		1.0150	1.0040	1.0000
Greece	1.0732	1.0063	0.9981	1.0000
Spain	1.0719	1.0326	1.0218	1.0000
France	1.1649	1.0778	1.0612	1.0000
Croatia			1.0237	1.0000
Italy	1.1123	1.0310	1.0310	1.0000
Cyprus	1.0718	1.0034	1.0248	1.0000
Latvia		1.0885	1.0546	1.0000
Lithuania			1.1497	1.0000
Luxembourg	1.1987	1.0968	1.0518	1.0000
Malta	1.1588	1.0845	1.0542	1.0000
Netherlands	1.2059	1.1057	1.0846	1.0000
Austria	1.2365	1.1236	1.0798	1.0000
Portugal	1.1265	1.0439	1.0135	1.0000
Slovenia	1.1397	1.0615	1.0549	1.0000
Slovakia	1.2164	1.1114	1.1054	1.0000
Finland	1.1597	1.0467	1.0318	1.0000

APPENDIX C. HOUSEHOLD BUDGET DECOMPOSITION

FIGURE C.1. Household budget decomposition and financial assets

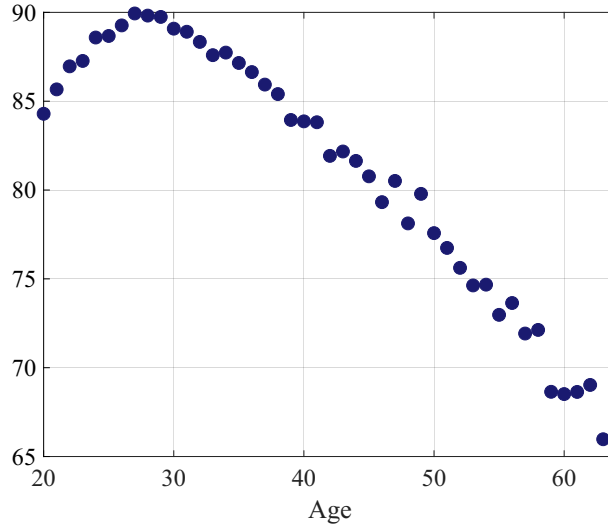


Notes: The figure plots the difference between the Paris Agreement and the Low Carbon Price scenarios for key components of the household budget: labor income, financial income, government transfers, and bequests. All variables are expressed in efficiency units per capita.

APPENDIX D. AGE PROFILE OF LABOR INCOME AS A SHARE OF TOTAL INCOME

Figure D.1 reports the share of labor income in total income across working-age cohorts, from age 20 to retirement at age 64, computed from the calibrated model at the pre-transition steady state. From age 20 to retirement at age 64, the share declines monotonically with age, reflecting the rising contribution of financial income over the working life. For the very young workers, the share initially rises with age: younger workers enter the labor market with lower remuneration (as shown in Figure 1), so their labor income contributes less to total resources at the very start of working life. The figure complements the results of Section 4 by documenting the strong life-cycle heterogeneity in households' exposure to the labor-income channel.

FIGURE D.1. Age profile of labor income as a share of total income (%)



Notes: The figure reports the age profile of labor income as a share of total income for working-age cohorts in the model's pre-transition steady state. The vertical axis measures shares in %. The horizontal axis reports working ages.

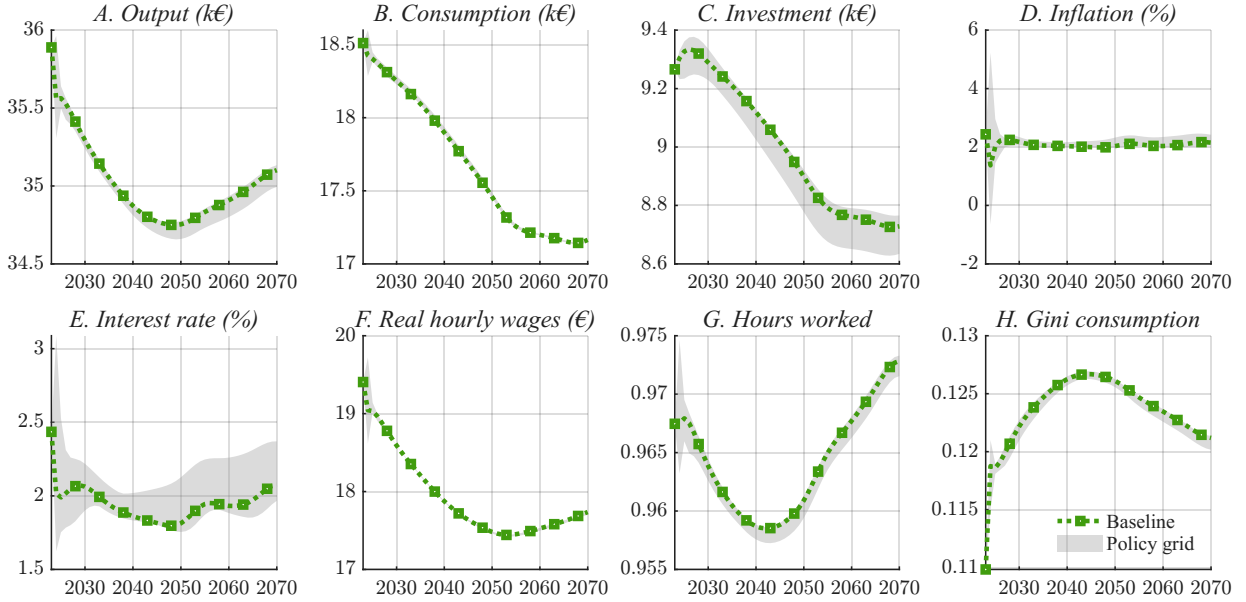
APPENDIX E. MONETARY POLICY AND THE FINANCIAL-WEALTH CHANNEL

The financial-wealth channel identified in Section 4 has a natural monetary dimension. During the transition, carbon pricing induces a period of disinflation. This raises the real value of nominal asset holdings and the return on financial wealth, generating a transfer toward older households, who are disproportionately exposed to financial income. A more aggressive monetary response could, in principle, dampen this effect by stabilizing inflation and limiting the initial revaluation of nominal assets. This section evaluates whether monetary policy can mitigate the resulting intergenerational redistribution.

E.1. Sensitivity to monetary policy rules. We first examine whether alternative monetary policy rules affect the transition dynamics. Figure E.1 reports equilibrium paths for a grid of Taylor-rule coefficients, with $\varphi_\pi \in [1.35, 5]$ and $\varphi_y \in [0, 1]$, and compares the resulting envelope with the baseline. The dispersion across rules is small for all main aggregates, including output, consumption, inflation, and inequality. Monetary policy therefore has only limited leverage over the aggregate dynamics of the transition: changing the inflation response mainly affects the timing of price adjustment, without generating sizable differences in real allocations or distributional outcomes.¹⁴

¹⁴This conclusion applies to the class of Taylor rules considered here. Under Ramsey-optimal monetary policy, the central bank could in principle exploit nominal rigidities more aggressively to influence real allocations. Solving for the Ramsey allocation, however, is computationally infeasible given the size of the model.

FIGURE E.1. Baseline transition and monetary policy grid

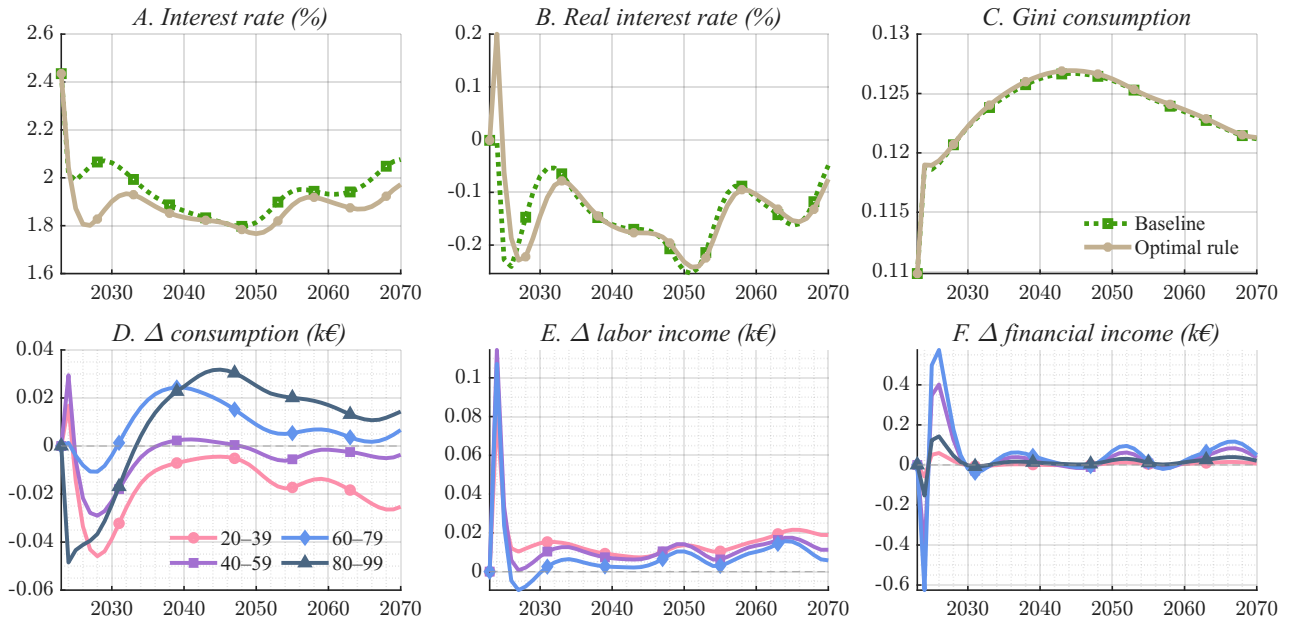


Notes: The figure displays macroeconomic transition paths under the baseline Taylor rule ($\varphi_\pi = 1.925$, $\varphi_y = 0.706$, dotted green line) alongside the policy grid envelope (grey area), which spans all simulated monetary rules over the parameter space $\varphi_\pi \in [1.35, 5]$ and $\varphi_y \in [0, 1]$. The shaded area represents the range of outcomes across the grid of alternative rules.

This limited role of monetary policy reflects two complementary factors. First, carbon pricing operates primarily through real forces: changes in production costs and factor returns. Nominal rigidities shape the short-run adjustment, but they do not drive the main redistributive effects. Second, the OLG structure weakens the standard intertemporal substitution channel. Because survival probabilities decline with age, households have shorter effective planning horizons, which dampens their consumption response to interest-rate changes. Together with borrowing constraints and portfolio adjustment costs, this attenuates the transmission from policy rates to saving, labor supply, wages, and ultimately labor income. As a result, monetary policy leaves the real redistribution generated by the green transition essentially unchanged.

E.2. Optimal monetary policy. We next consider a stronger monetary policy response within the specified class of rules. We select the policy coefficients that maximize mean aggregate welfare over 2026–2050 among the parameter combinations considered on the simulated grid. The resulting optimal rule responds more strongly to inflation and places no weight on output growth, with $\varphi_\pi = 2.65$ and $\varphi_y = 0$. Figure E.2 compares the transition dynamics under

FIGURE E.2. Baseline versus optimal monetary policy during the transition



Notes: The upper panels display the transition dynamics under the baseline Taylor rule ($\varphi_\pi = 1.925$, $\varphi_y = 0.706$) and under the optimal (welfare-maximizing) monetary policy rule ($\varphi_\pi = 2.65$, $\varphi_y = 0$). The lower panels display the difference in consumption, labor income, and financial income between the two rules for each household cohort.

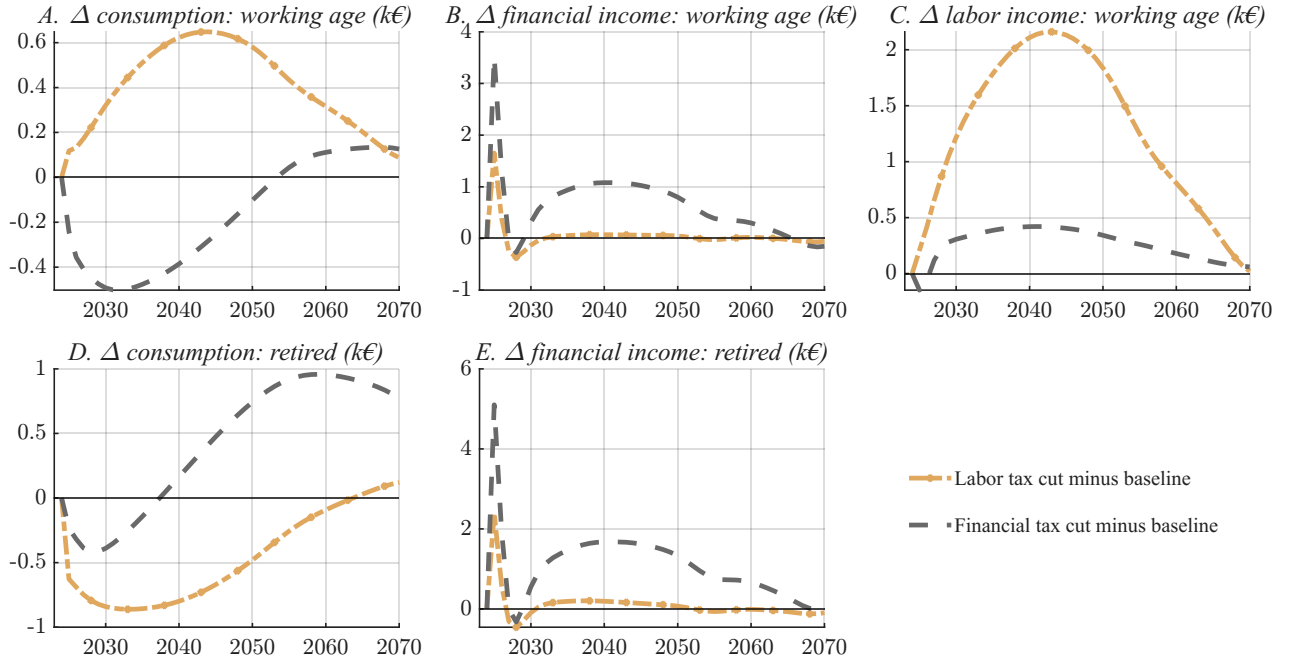
the optimal rule with those obtained under the baseline specification. Despite the more aggressive response to inflation, the resulting allocation remains close to the baseline. Aggregate quantities, inflation dynamics, and inequality therefore exhibit only minor deviations across the two regimes. The lower panels confirm this result at age-group level. Differences in consumption, labor income, and financial income are concentrated in the first period of the transition and dissipate rapidly thereafter, remaining quantitatively negligible for all age groups beyond the initial adjustment. Even under the welfare-maximizing rule, monetary policy leaves the redistribution force largely unaffected.

Within the class of Taylor rules and welfare criterion considered here, the simulations indicate limited scope for monetary policy to alter distributional outcomes. Even under an optimal policy rule, the intergenerational redistribution remains largely intact. This contrasts with fiscal policy, which can directly target the underlying sources of redistribution by modifying after-tax income and asset returns. As a result, fiscal instruments, rather than monetary policy, are the primary tool for addressing the distributional effects of carbon pricing.

APPENDIX F. DISTRIBUTIONAL CHANNELS OF CARBON REVENUE RECYCLING

Figure F.1 documents the distributional channels underlying the two recycling schemes considered in Section 6. Labor-income tax relief primarily raises labor income and consumption among working-age households during the costly phase of the transition, while providing little direct support to retirees. Financial-income tax relief instead raises financial income, with substantially larger effects for retirees because of their greater asset holdings. These differences in income exposure translate into the contrasting welfare incidence across age groups documented in the main text. They illustrate how the choice of recycling instrument reshapes the redistribution force even when both schemes generate aggregate welfare gains.

FIGURE F.1. Distributional channels of alternative carbon revenue recycling schemes



Notes: The panels show the difference (Δ) between the path of a variable (consumption, labor income and financial income) obtained under a given revenue recycling scheme and that obtained under the baseline. The first row (Panels A–C) displays the results for the working age population (aged 20–64) and the second row (Panels D–E) for the retired (aged 65–99). The baseline corresponds to the Paris-Agreement scenario, in which carbon revenue is redistributed to households through lump sum transfers.

APPENDIX G. ROBUSTNESS OF THE POLICY RANGE

The policy range identified in Section 6 rests on specific assumptions about nominal rigidities, the carbon-price trajectory, and demographic structure. We assess its robustness across six alternative specifications. In each case, we recompute the range under the same two distributional requirements: non-negative CEV for retirees and no increase in the consumption Gini relative to the lump-sum baseline. As in the main text, both measures are evaluated at each date and then averaged over the full transition horizon. Figure G.1 summarizes the results.

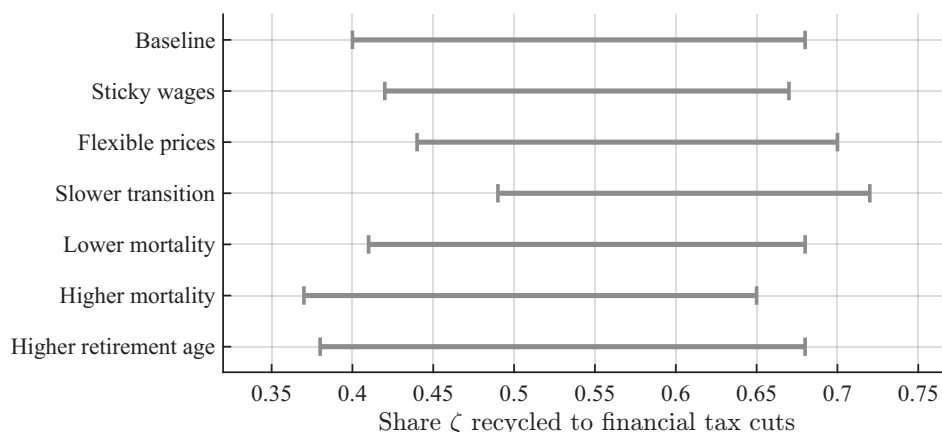
Nominal rigidities. In the first experiment, we introduce sticky wages. Each household supplies a differentiated labor type, aggregated through a constant-elasticity-of-substitution aggregator. Wage adjustment is subject to Rotemberg costs, $\frac{\kappa_w}{2} \left(\frac{W_t}{W_{t-1}(1+g_{Z,t})} \pi_t - \pi \right)^2 H_t$ with κ_w calibrated so that the slope of the wage Phillips curve matches that of the price Phillips curve. The policy range narrows slightly to approximately [0.42, 0.67]. In the second experiment, we remove price stickiness entirely by setting $\kappa_p = 0$. The range shifts to approximately [0.44, 0.70]. These experiments indicate that the existence of the policy range is not driven by the baseline specification of nominal rigidities.

Transition speed. In the third experiment, we replace the Paris-Agreement carbon-price trajectory with the socially optimal path from Barrage and Nordhaus (2024) under a 2 percent rate of time preference. The resulting carbon price rises more gradually, altering both the timing of transition costs and the revenues available for recycling. The policy range shifts rightward to approximately [0.49, 0.72]. A larger share of revenues must therefore be directed toward financial-income tax relief to compensate retirees, while the inequality constraint becomes less restrictive.

Demographic structure. We next perturb the mortality schedule used to calibrate cohort survival probabilities. Using United Nations World Population Prospects mortality rates for 2033 (lower mortality) yields a range of approximately [0.41, 0.68]. A larger working-age cohort amplifies the labor-income channel, making the inequality constraint bind more tightly and raising $\zeta_{\min}$, while the smaller retiree share limits the redistribution required through financial tax cuts, pulling $\zeta_{\max}$ inward. Using realized mortality rates for 2003 instead (higher mortality) shifts the range leftward to approximately [0.37, 0.65]. The quantitative boundaries therefore respond to the age composition of the population, but the policy range remains non-empty under both alternative demographic structures.

Retirement age. In the final experiment, we raise the mandatory retirement threshold from 64 to 65, reallocating a marginal cohort from retirement to employment along the transition.

FIGURE G.1. Robustness of the policy range across model variants



Notes: Each bar reports the policy range $[\zeta_{\min}, \zeta_{\max}]$ for a given model specification, where ζ is the share of carbon revenues recycled through financial income tax cuts. The range is defined as the set of ζ values satisfying both distributional requirements: non-negative consumption-equivalent welfare gains for retirees and no increase in the consumption Gini relative to the lump-sum baseline.

The policy range shifts to approximately $[0.38, 0.68]$: the lower bound falls slightly below its baseline value, while the upper bound is essentially unchanged. Hence, the result is also robust to a marginal extension of working life.

Across all six specifications, the policy range remains non-empty. Its lower bound varies between 0.37 and 0.49 and its upper bound between 0.65 and 0.72, compared with $[0.40, 0.68]$ in the baseline. Some financial-income tax relief is therefore required in every specification, while excessive reliance on it eventually violates the inequality criterion. The quantitative boundaries vary with the model specification, but the central trade-off underlying the distributional third dividend remains unchanged.